

7-ADIC GALOIS REPRESENTATIONS OF ELLIPTIC CURVES OVER THE RATIONALS VIA KUMMER DESCENT

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ABSTRACT. We show that the two modular curves of level 49 and genus 9 left open by the work of Rouse, Sutherland and Zureick-Brown, and of Furio and Lombardo, have no non-CM rational points. Together with the theorem of Furio and Lombardo on $X_{\text{ns}}^+(49)$, this completes the classification of the 7-adic images of Galois of non-CM elliptic curves over $\mathbb{Q}$. The proof is a Kummer descent on the superelliptic equations $F(x, y) = kw^7$ of Furio and Lombardo, in the cases $7 \mid k$ that they left open. The covering curves are twists of the Fermat septic, they map to twists of the Klein quartic, and a 7-adic computation shows that the twist attached to a solution is trivial. The rational points of the Klein quartic were determined by Hurwitz, who reduced the question to Fermat's Last Theorem for exponent 7, proved by Lamé. Thus, the last open case of the 7-adic part of Mazur's Program B rests on Fermat's Last Theorem for exponent 7.

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1. INTRODUCTION

Let E be an elliptic curve over $\mathbb{Q}$ without complex multiplication and let ℓ be a prime. By Serre's open image theorem [Ser72], the ℓ -adic Galois representation ρ_{E, ℓ^∞} of $\text{Gal}_{\mathbb{Q}}$ has open image in $\text{GL}_2(\mathbb{Z}_\ell)$, and Mazur's *Program B* [Maz77, p. 109] asks for the classification of the subgroups that occur. Rouse, Sutherland and the second author [RSZB22, Theorem 1.6] carried out this classification up to the determination of the rational points of a short list of modular curves (we give their LMFDB labels [LMFDB26]): the curves $X_{\text{ns}}^+(\ell)$ attached to the normalizer of a non-split Cartan subgroup, for primes $\ell > 17$; the four curves $X_{\text{ns}}^+(27) = 27.243.12.a.1$, $X_{\text{ns}}^+(25) = 25.250.14.a.1$, $X_{\text{ns}}^+(49) = 49.1029.69.a.1$ and $X_{\text{ns}}^+(121) = 121.6655.511.a.1$; and two curves of level 49 and genus 9, which, following Furio and Lombardo [FL26, Definition 2.7, Remark 2.8], we denote by $X_{\text{ns}}^\sharp(49) = 49.147.9.a.1$ and $X_{\text{sp}}^\sharp(49) = 49.196.9.a.1$. The curve $X_{\text{ns}}^+(27)$ has since been settled by Balakrishnan, Betts,

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Hast, Jha and Müller [BBH⁺25], and $X_{\text{ns}}^+(49)$ by Furio and Lombardo, who proved that it has exactly seven rational points, all of them CM [FL26, Theorem 1.4]. Our work here builds on that of Furio and Lombardo, and our main result is the computation of the rational points on the curves $X_{\text{ns}}^\sharp(49)$ and $X_{\text{sp}}^\sharp(49)$, thereby completing the classification of 7-adic images of Galois for non-CM elliptic curves over $\mathbb{Q}$.

Furio and Lombardo reduced the computation of the rational points on $X_{\text{ns}}^\sharp(49)$ and $X_{\text{sp}}^\sharp(49)$ to the determination of the rational points of a single plane quartic $X_{E_3}(7)$, a twist of the Klein quartic whose Jacobian has Mordell–Weil rank 3, and they conjectured that this quartic has exactly four rational points [FL26, Conjecture 1.6]. We do not address this conjecture: our proof is unconditional and takes a different route.

Remark 1.1. On September 18, 2026, Eray Karabiyik communicated to us that he was able to prove [FL26, Conjecture 1.6] directly [Kar26]. Karabiyik’s proof, in essence, does quadratic Chabauty over number fields by using an elliptic quotient over a number field of degree 21.

Remark 1.2. On September 28, 2026, Nguyen Xuan Tho emailed us a preprint [Xua26] with an AI-generated proof of [FL26, Conjecture 1.6].

The Diophantine statement. The curves $X_{\text{ns}}^\sharp(49)$ and $X_{\text{sp}}^\sharp(49)$ admit natural maps of degree 7 to the genus zero curves $X_{\text{ns}}^+(7) = \textcolor{red}{7.21.0.a.1}$ and $X_{\text{sp}}^+(7) = \textcolor{red}{7.28.0.a.1}$ attached to the normalizers of a non-split and of a split Cartan subgroup of $\text{GL}_2(\mathbb{F}_7)$ (the precise definitions are recalled in Section 2.2). The j -maps of $X_{\text{ns}}^+(7)$ and $X_{\text{sp}}^+(7)$, in Zywin’s coordinates (the functions J_6 and J_2 of [Zyw15, Section 1.4]; see [Zyw15, Theorem 1.5(ii)] and [FL26, Theorem 2.1]), have denominators $f_{\text{ns}}(\mathbf{t})^7$ and $f_{\text{sp}}(\mathbf{t})^7$, where

$$(1.1) \quad f_{\text{ns}}(\mathbf{t}) := \mathbf{t}^3 - 7\mathbf{t}^2 + 7\mathbf{t} + 7, \quad f_{\text{sp}}(\mathbf{t}) := \mathbf{t}^3 - 4\mathbf{t}^2 + 3\mathbf{t} + 1.$$

Both cubics define the field $K := \mathbb{Q}(\zeta_7)^+$. Let $F(\mathbf{x}, \mathbf{y})$ denote the homogenization of $f(\mathbf{t})$:

$$(1.2) \quad F_{\text{ns}}(\mathbf{x}, \mathbf{y}) := \mathbf{x}^3 - 7\mathbf{x}^2\mathbf{y} + 7\mathbf{x}\mathbf{y}^2 + 7\mathbf{y}^3, \quad F_{\text{sp}}(\mathbf{x}, \mathbf{y}) := \mathbf{x}^3 - 4\mathbf{x}^2\mathbf{y} + 3\mathbf{x}\mathbf{y}^2 + \mathbf{y}^3.$$

Furio and Lombardo showed [FL26, Proposition 3.1] that a non-CM rational point on $X_{\text{ns}}^\sharp(49)$ or $X_{\text{sp}}^\sharp(49)$ yields coprime integers x, y and an integer w such that $F(x, y) = kw^7$, with $k \in \{7, 56\}$ in the non-split case and $k \in \{1, 7\}$ in the split case; we rederive this in Section 2. Our main result is the solution of these equations when $7 \mid k$.

Theorem 1. *Let (F, k) be one of the pairs $(F_{\text{ns}}, 7)$, $(F_{\text{ns}}, 56)$, $(F_{\text{sp}}, 7)$. The integer solutions (x, y, w) of*

$$(1.3) \quad F(\mathbf{x}, \mathbf{y}) = k \mathbf{w}^7, \quad \gcd(x, y) = 1,$$

all satisfy $w = \pm 1$. Moreover, they are exactly the ones listed in Table 1.

For the remaining pairs, those with $7 \nmid k$, the analogous statement is a theorem of Furio and Lombardo: every solution of $F_{\text{ns}}(x, y) = kw^7$ with $k \in \{1, 8\}$ and of $F_{\text{sp}}(x, y) = w^7$ in coprime integers has $w = \pm 1$ [FL26, Corollary 3.3(1), Theorem 3.5(1)]. We use this in the split case of the next theorem, where $k = 1$ is allowed. The cases $7 \mid k$ are precisely the ones that lead Furio and Lombardo to the twist of the Klein quartic of rank 3.

Theorem 2. *Every rational point of $X_{\text{ns}}^\sharp(49)$ and of $X_{\text{sp}}^\sharp(49)$ is a cusp or a CM point. More precisely, every non-cuspidal rational point of either curve has j -invariant 0. In particular,*

(F, k)	$(F_{\text{ns}}, 7)$	$(F_{\text{ns}}, 56)$		$(F_{\text{sp}}, 7)$		
(x, y, w)	$\pm(0, 1, 1)$	$\pm(7, 1, 1)$	$\pm(7, 3, -1)$	$\pm(1, -1, 1)$	$\pm(5, 2, -1)$	$\pm(4, 3, 1)$
curve	$X_{\text{ns}}^+(7)(\mathbb{Q})$			$X_{\text{sp}}^+(7)(\mathbb{Q})$		
$t = x/y$	0	7	7/3	-1	5/2	4/3
$j(t)$	0	non-CM	non-CM	0	non-CM	non-CM

TABLE 1. The solutions of (1.3) for the three pairs (F, k) with $7 \mid k$ (Theorem 1), the corresponding rational points $t = x/y$ of $X_{\text{ns}}^+(7)$ and $X_{\text{sp}}^+(7)$, and whether their j -invariants are CM.

the four non-CM points of Table 1 do not lift to rational points of $X_{\text{ns}}^\sharp(49)$, respectively of $X_{\text{sp}}^\sharp(49)$.

Corollary 3. *Theorem 1.7 of [FL26] holds unconditionally. In particular, for every elliptic curve E over $\mathbb{Q}$ without complex multiplication, the 7-adic image $G = \rho_{E,7^\infty}(\text{Gal}_{\mathbb{Q}})$ satisfies one of the first two alternatives of [RSZB22, Theorem 1.6]: either X_G has infinitely many rational points and G is one of the groups of Sutherland and Zywina [SZ17], or G appears in [RSZB22, Table 1].*

Proof. Alternative (iii) of [RSZB22, Theorem 1.6] at level 7^2 is excluded by [FL26, Theorem 1.4], and alternative (iv) by Theorem 2. $\square$

Summary of the proofs. Furio and Lombardo attach to a solution of $F(x, y) = kw^7$ a solution of a generalized Fermat equation of signature $(2, 3, 7)$, by means of the covariants of the cubic form F [FL26, Proposition 3.2, Corollary 3.3] (compare Bennett and Dahmen [BD13, Equation (4)]), and then apply the modular method of Poonen, Schaefer and Stoll [PSS07, Section 4]: the solutions are governed by rational points on twists $X_E(7)$ of the Klein quartic [HK03]. When $7 \nmid k$ the equation is $a^2 + 28b^3 = 27c^7$ and all the twists that occur have rank at most 2; when $7 \mid k$ it is $a^2 + 196b^3 = 27c^7$, and one twist of rank 3 remains.

We instead descend directly on the superelliptic equation, in the style of Kummer [DG95, Section 2.1]. This method is streamlined and explained by Bruin in [Bru02, Section 3.1], [Bru06, Section 3]. The argument has six steps. Along the way, the set of classes by which we have to twist shrinks: from the Selmer group $K(7, \mathcal{S})$ to a subset Δ of 49 classes in step (2), and to 7 of these in step (4).

- (1) **The equations (Section 2).** A rational point of $X_{\text{ns}}^\sharp(49)$ or $X_{\text{sp}}^\sharp(49)$ lies over a point $t = x/y$ of the genus zero curve $X_{\text{ns}}^+(7)$, respectively $X_{\text{sp}}^+(7)$, and the exponent 7 in $F(x, y) = kw^7$ comes from the primes of potentially multiplicative reduction. At such a prime the Tate curve makes inertia act unipotently, and the shape of the level 49 group forces the valuation of $F(x, y)$ to be divisible by 7. This recovers [FL26, Proposition 3.1]. In the split case the argument of [FL26] uses an element of the decomposition group that does not exist at the primes p with $p^6 \equiv 1 \pmod{49}$; the proof given here uses only inertia, together with the position of the rational cusp (Lemma 2.5, Remark 2.7).

- (2) **Kummer descent to twists of the Fermat septic (Section 3).** Let $\theta \in K$ be a root of f . Over K the form $F(x, y)$ is the norm of $x - \theta y$, and since K has class number one, a solution (x, y, w) to (1.3) yields $x - \theta y = \delta \omega^7$ with $\omega \in K^\times$ and δ an $\mathcal{S}$ -unit, i.e., a unit outside the set $\mathcal{S}$ of primes dividing $k \operatorname{disc}(f)$. So $x - \theta y$ is a seventh power in K up to a factor δ that only matters through its class in the Selmer group $K(7, \mathcal{S}) = \mathcal{O}_{K, \mathcal{S}}^\times / \mathcal{O}_{K, \mathcal{S}}^{\times 7}$, and comparing norms restricts this class to an explicit subset $\Delta(\theta, k)$ of 49 classes (Proposition 3.3). Each $\delta \in \Delta(\theta, k)$ gives a plane curve, as follows. Write $\omega = a_0 + a_1\theta + a_2\theta^2$ with $a_n \in \mathbb{Q}$, and put $\mathbf{a} = (a_0, a_1, a_2)$. There are homogeneous ternary forms $P_{\theta, \delta}^{(0)}, P_{\theta, \delta}^{(1)}, P_{\theta, \delta}^{(2)}$ of degree 7 with rational coefficients such that

$$\delta (a_0 + a_1\theta + a_2\theta^2)^7 = P_{\theta, \delta}^{(0)}(\mathbf{a}) + P_{\theta, \delta}^{(1)}(\mathbf{a})\theta + P_{\theta, \delta}^{(2)}(\mathbf{a})\theta^2.$$

Since $x - \theta y$ has no θ^2 term, the point $(a_0 : a_1 : a_2)$ lies on the plane septic $C_{\theta, \delta} : P_{\theta, \delta}^{(2)} = 0$ over $\mathbb{Q}$, of genus 15, and $(x : y) = (P_{\theta, \delta}^{(0)}(\mathbf{a}) : -P_{\theta, \delta}^{(1)}(\mathbf{a}))$. So the rational points of the curves $C_{\theta, \delta}$ account for all the solutions. Over K each of these curves becomes a diagonal septic, a twist of the Fermat curve $\mathcal{F} : u_0^7 + u_1^7 + u_2^7 = 0$ to which it is isomorphic over $\overline{\mathbb{Q}}$. Remarkably, all six pairs (F, k) that arise in Section 2 lead to the same 49 twists of the Fermat septic.

- (3) **Quotient to twists of the Klein quartic (Section 4).** The Fermat septic $\mathcal{F}$ is a cyclic unramified cover of degree 7 of the Klein quartic curve $\mathcal{Z} : v_0^3v_1 + v_1^3v_2 + v_2^3v_0 = 0$ [Elk99, Section 3.2]. This cover survives the twisting: each of our septics $C_{\theta, \delta}$ maps, over $\mathbb{Q}$, onto a twist $Z_{\theta, \delta}$ of the Klein quartic, so it is enough to find the rational points of these plane quartics. Which twist occurs is recorded by the class of $\delta/f'(\theta)$ in the group $K^\times/\mathbb{Q}^\times(K^\times)^{3+\sigma}$ (Lemma 4.3). For $\delta \in \Delta(\theta, k)$ this class lies in a subgroup of order 7, so the 49 septics of step (2) map to at most 7 quartics up to isomorphism, and in fact to only three (Proposition 5.6).
- (4) **The 7-adic step (Section 5).** When $7 \mid F(x, y)$, a short computation in the completion of K at 7 shows that this class vanishes (Proposition 5.3): a homomorphism κ to $\mathbb{F}_7$ must vanish on it. This rules out 42 of the 49 classes $\delta \in \Delta(\theta, k)$, and the remaining 7 all give quartics $\mathbb{Q}$ -isomorphic to one fixed quartic Z_0 . A classical change of coordinates of Klein, defined over K [Kle78, §5], [Elk99, (1.3)], gives an isomorphism over $\mathbb{Q}$ between Z_0 and the Klein quartic itself (Proposition 4.4, Corollary 4.7). Conversely, for the other 42 classes the quartic $Z_{\theta, \delta}$ is not isomorphic to the Klein quartic over $\mathbb{Q}$ (Proposition 5.6). The cases $7 \mid k$ are exactly those that were missing from [FL26].
- (5) **Hurwitz (Lemma 4.5).** The Klein quartic has only its three obvious rational points: Hurwitz [Hur08] showed that any other would give a nontrivial solution of Fermat's equation of exponent 7, and Elkies [Elk99, Section 3.1] gives a direct proof through an elliptic quotient of rank zero. Pulling the three points back gives the solutions listed in Theorem 1. So, via Hurwitz, the last open case of the 7-adic Program B rests on Fermat's Last Theorem for exponent 7.
- (6) **From Theorem 1 to Theorem 2 (Section 6).** Theorem 1, together with the result of Furio and Lombardo for $k = 1$ recalled before Theorem 2, leaves eight non-CM j -invariants of points of $X_{\text{ns}}^+(7)$ or $X_{\text{sp}}^+(7)$ that could still lift to a rational point of $X_{\text{ns}}^\sharp(49)$ or $X_{\text{sp}}^\sharp(49)$ (Table 4). One could rule this out using the equations for the maps $X_\bullet^\sharp(49) \rightarrow$

$X_{\bullet}^+(7)$; instead we use the following group-theoretic criterion (Corollary 6.2). Let E be an elliptic curve over $\mathbb{Q}$ whose mod-49 Galois image is contained, up to conjugacy, in one of the groups $G_{\text{ns}}^{\sharp}(49)$ or $G_{\text{sp}}^{\sharp}(49)$ defining these two curves (Definition 2.3), and let $p \neq 7$ be a prime of good reduction, with trace of Frobenius a_p . If $7 \nmid a_p(a_p^2 - 4p)$, then Frob_p^{48} acts on $E[49](\overline{\mathbb{Q}})$ as a scalar. Indeed, modulo 7 such a Frobenius lies in the Cartan subgroup, so its 48th power is 1 modulo 7, and the shape of the level 49 group forces this power to be scalar modulo 49. For each of the eight candidates, the criterion fails at a small prime p . The CM points are handled by the same criterion, together with [FL26, Lemma 2.11] in the non-split case.

The relevance of quotients of Fermat curves to descent problems of this kind goes back at least to Faddeev [Fad61], Gross and Rohrlich [GR78], McCallum [McC88], and McCallum and Tzermias [MT03]. Twists of the Fermat septic covering twists of the Klein quartic also appear in [PSS07, Section 10.1]. Figure 1 summarizes the curves and maps involved.

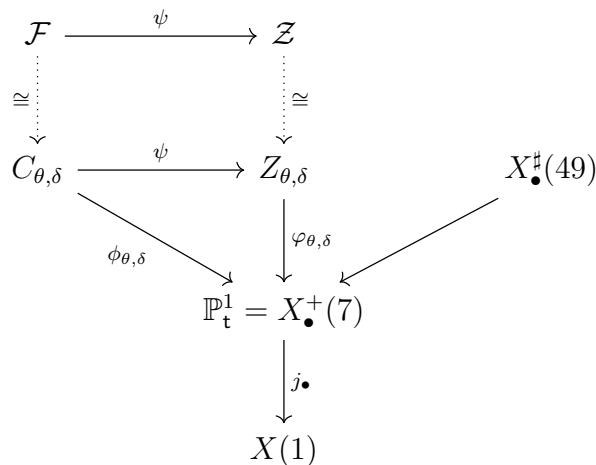


FIGURE 1. The curves of the proof, for one $\delta \in \Delta(\theta, k)$. The two vertical isomorphisms (dotted) are defined over $\overline{\mathbb{Q}}$, everything else over $\mathbb{Q}$. The maps ψ and $\varphi_{\theta,\delta}$ have degree 7, and $\phi_{\theta,\delta}$ has degree 49 and is branched exactly over the three roots of f , which are cusps of $X_{\bullet}^+(7)$. The covering $X_{\bullet}^{\sharp}(49) \rightarrow X_{\bullet}^+(7)$ has degree 7, and $j_{\bullet}$ has degree 21 for $\bullet = \text{ns}$ and 28 for $\bullet = \text{sp}$. A solution (x, y, w) of $F(x, y) = kw^7$ is a rational point $(x : y)$ of $\mathbb{P}_t^1$ in the image of $\phi_{\theta,\delta}$ for some $\delta \in \Delta(\theta, k)$. When $7 \mid k$, every solution comes from a δ with $\kappa(\delta/f'(\theta)) = 0$; these are exactly the δ for which $Z_{\theta,\delta}$ is isomorphic to $\mathcal{Z}$ over $\mathbb{Q}$ (Proposition 5.3, Corollary 4.7, Proposition 5.6), and the three rational points of that quartic map to $t \in \{0, 7, 7/3\}$ for $\bullet = \text{ns}$ and to $t \in \{-1, 5/2, 4/3\}$ for $\bullet = \text{sp}$, the points of Theorem 1.

Code. Every computational claim in this paper is verified by Magma [BCP97] code. The code and the source of this paper are available at <https://github.com/sarangop1728/7-adic-settlers-of-cartan>. Both are generated from one literate file, in which each statement of the paper is followed by the code that verifies it. There is one script for each of Sections 2 to 6, and a script `descent.m` that runs the descent of Sections 3 to 5

from the Selmer sets to Table 1; each runs in under a minute. A web version of the paper, at <https://sarangop1728.github.io/7-adic-settlers-of-cartan/>, shows the code and its output below the statements it verifies.

AI use disclosure. Throughout this project we used AI models (GPT-5.6 Sol, and Claude Opus 5, Opus 5.5, and Fable 5.1) for mathematical exploration, to find simplifications of our arguments, to write and run **Magma** code, and for copy-editing. The origin of the main ideas and the precise mathematical use of AI are described in more detail in Section 7. All mathematical results in this paper, and the code and computations verifying them, have been thoroughly vetted by the authors, who take full responsibility for the content of this work.

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Notation. Throughout, ζ_7 is a fixed primitive seventh root of unity, $K = \mathbb{Q}(\zeta_7)^+$ is the totally real cyclic cubic field of discriminant 49, which has class number one, and σ is the generator of $\text{Gal}(K/\mathbb{Q})$ induced by $\zeta_7 \mapsto \zeta_7^2$. We write Tr and Norm for the trace and norm of $K/\mathbb{Q}$. Lower-case sans-serif letters $x, y, w, t, \dots$ are polynomial variables, and the corresponding italic letters are their values. We write f for one of the cubics in (1.1), F for its homogenization, and $\theta \in K$ for the following root of f :

$$(1.4) \quad \theta_{\text{sp}} := 1 - (\zeta_7 + \zeta_7^{-1}), \quad \theta_{\text{ns}} := 3 + 2(\zeta_7 + \zeta_7^{-1}) = 5 - 2\theta_{\text{sp}}.$$

One has $\sigma(\theta_{\text{sp}}) = -\theta_{\text{sp}}^2 + 2\theta_{\text{sp}} + 2$, the ring of integers of K is $\mathcal{O}_K = \mathbb{Z}[\theta_{\text{sp}}]$, and $\mathbb{Z}[\theta_{\text{ns}}]$ has index 8 in $\mathcal{O}_K$. The prime 2 is inert in K and 7 is totally ramified. Since $\theta_{\text{ns}} - \sigma^i(\theta_{\text{ns}}) = -2(\theta_{\text{sp}} - \sigma^i(\theta_{\text{sp}}))$ and $f'(\theta) = \prod_{i=1,2}(\theta - \sigma^i(\theta))$,

$$(1.5) \quad f'_{\text{ns}}(\theta_{\text{ns}}) = 4f'_{\text{sp}}(\theta_{\text{sp}}), \quad \text{Norm}(f'_{\text{sp}}(\theta_{\text{sp}})) = -\text{disc}(f_{\text{sp}}) = -7^2.$$

2. THE EQUATIONS

This section rederives [FL26, Proposition 3.1], the passage from rational points on the modular curves $X_{\text{ns}}^\sharp(49)$ and $X_{\text{sp}}^\sharp(49)$ to the superelliptic equations.

2.1. Zywin's models. By [Zyw15, Section 1.4, Theorem 1.5(ii)], where j_{ns} and j_{sp} below are the functions J_6 and J_2 (see also [FL26, Theorem 2.1]), the j -maps of $X_{\text{ns}}^+(7) \cong \mathbb{P}_t^1$ and $X_{\text{sp}}^+(7) \cong \mathbb{P}_t^1$ are given by the rational functions

$$(2.1) \quad j_{\text{ns}}(t) = \frac{64t^3(t^2 + 7)^3(t^2 - 7t + 14)^3(5t^2 - 14t - 7)^3}{(t^3 - 7t^2 + 7t + 7)^7},$$

$$(2.2) \quad j_{\text{sp}}(t) = \frac{t(t+1)^3(t^2 - 5t + 1)^3(t^2 - 5t + 8)^3(t^4 - 5t^3 + 8t^2 - 7t + 7)^3}{(t^3 - 4t^2 + 3t + 1)^7}.$$

The first has degree 21 and its three poles, the roots of f_{ns} , are the cusps of $X_{\text{ns}}^+(7)$. The second has degree 28; it has poles of order 7 at the roots of f_{sp} and at $t = \infty$, the four

cusps of $X_{\text{sp}}^+(7)$, of which only $t = \infty$ is rational. Homogenizing, $j_{\text{ns}} = H_{\text{ns}}^3/F_{\text{ns}}^7$ and $j_{\text{sp}} = xH_{\text{sp}}^3/(yF_{\text{sp}})^7$, where F_{ns} and F_{sp} are the cubic forms (1.2) and

$$H_{\text{ns}}(x, y) := 4x(x^2 + 7y^2)(x^2 - 7xy + 14y^2)(5x^2 - 14xy - 7y^2),$$

$$H_{\text{sp}}(x, y) := (x + y)(x^2 - 5xy + y^2)(x^2 - 5xy + 8y^2)(x^4 - 5x^3y + 8x^2y^2 - 7xy^3 + 7y^4).$$

The following elementary facts about these forms are used throughout.

Lemma 2.1. *Let x, y be coprime integers.*

- (1) *The greatest common divisor of $F_{\text{ns}}(x, y)$ and $H_{\text{ns}}(x, y)$ divides $2^{21}7^7$, and that of $xH_{\text{sp}}(x, y)$ and $yF_{\text{sp}}(x, y)$ divides 7^7 . Moreover $\gcd(y, F(x, y)) = 1$ for both forms.*
- (2) *$v_7(F(x, y)) \leq 1$ for both forms. Moreover:*
 - (a) *$7 \mid F_{\text{ns}}(x, y)$ if and only if $7 \mid x$;*
 - (b) *$7 \mid F_{\text{sp}}(x, y)$ if and only if $7 \mid x + y$.*
- (3) *$v_2(F_{\text{ns}}(x, y)) \in \{0, 3\}$, with value 3 exactly when x and y are both odd; and $F_{\text{sp}}(x, y)$ is odd.*
- (4) *$F_{\text{ns}}(x, y) = F_{\text{sp}}(x - 5y, -2y)$.*

Proof. One computes the resultants $\text{res}_x(F_{\text{ns}}, H_{\text{ns}}) = \pm 2^{21}7^7y^{21}$ and $\text{res}_x(F_{\text{sp}}, H_{\text{sp}}) = \pm 7^7y^{27}$. Both forms F are congruent to x^3 modulo y , and $H_{\text{sp}} \equiv x^9$ modulo y , while $F_{\text{sp}} \equiv y^3$ modulo x ; this gives the coprimality statements and (1). Parts (2) and (3) are congruence computations. Call a pair $(x, y) \in (\mathbb{Z}/n\mathbb{Z})^2$ **primitive** if it is not congruent to $(0, 0)$ modulo any prime $p \mid n$; the reduction of a coprime pair of integers is primitive. Then $F(x, y) \equiv 0 \pmod{49}$ and $F_{\text{ns}}(x, y) \equiv 0 \pmod{16}$ have no primitive solutions, $F_{\text{ns}} \equiv x^3$ and $F_{\text{sp}} \equiv (x + y)^3 \pmod{7}$, $F_{\text{ns}} \equiv (x + y)^3 \pmod{2}$, and $F_{\text{sp}}(x, y) \equiv 1 \pmod{2}$ for the three primitive pairs $(1, 0)$, $(0, 1)$, $(1, 1)$ modulo 2. That $v_2(F_{\text{ns}}(x, y))$ is a multiple of 3 also follows from the fact that 2 is inert in K . Part (4) holds because $\theta_{\text{ns}} = 5 - 2\theta_{\text{sp}}$: indeed $x - \theta_{\text{ns}}y = (x - 5y) - \theta_{\text{sp}}(-2y)$. $\square$

2.2. The groups. We recall [FL26, Definitions 2.6 and 2.7]. Let $\varepsilon = 3$, the least quadratic non-residue modulo 7. For $\bullet \in \{\text{ns}, \text{sp}\}$, Table 2 lists the Cartan subalgebra $\mathfrak{c}_\bullet$ of $M_2(\mathbb{F}_7)$, the Cartan subgroup $C_\bullet(7) = \mathfrak{c}_\bullet \cap \text{GL}_2(\mathbb{F}_7)$ and its normalizer $C_\bullet^+(7)$ in $\text{GL}_2(\mathbb{F}_7)$, and a subspace $V_\bullet$ of $M_2(\mathbb{F}_7)$. Let $N := I + 7M_2(\mathbb{F}_7)$ be the kernel of reduction $\text{GL}_2(\mathbb{Z}/49) \rightarrow \text{GL}_2(\mathbb{F}_7)$. Whenever $\bullet$ is fixed, as in the rest of this section, in Proposition 2.6 and in Lemma 6.1, we drop it and the level from the notation, and write $\mathfrak{c}$, C , C^+ , V and $G^\sharp$ for $\mathfrak{c}_\bullet$, $C_\bullet(7)$, $C_\bullet^+(7)$, $V_\bullet$ and $G_\bullet^\sharp(49)$.

$\bullet$	$\mathfrak{c} = \mathfrak{c}_\bullet$	$C = C_\bullet(7)$	$C^+ = C_\bullet^+(7)$	$V = V_\bullet$
ns	$\begin{pmatrix} a & \varepsilon b \\ b & a \end{pmatrix}$	$\begin{pmatrix} a & \varepsilon b \\ b & a \end{pmatrix}$	$C_{\text{ns}}(7) \cup \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} C_{\text{ns}}(7)$	$\begin{pmatrix} a & \varepsilon b \\ -b & d \end{pmatrix}$
sp	$\begin{pmatrix} a & 0 \\ 0 & d \end{pmatrix}$	$\begin{pmatrix} a & 0 \\ 0 & d \end{pmatrix}$	$C_{\text{sp}}(7) \cup \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} C_{\text{sp}}(7)$	$\begin{pmatrix} a & b \\ c & a \end{pmatrix}$

TABLE 2. The Cartan data at level 7. Here, $C_\bullet(7) \subset \text{GL}_2(\mathbb{F}_7)$ and $V_\bullet \subset M_2(\mathbb{F}_7)$.

The groups $G^\sharp$ of Furio and Lombardo are the subgroups of $\text{GL}_2(\mathbb{Z}/49)$ that reduce onto C^+ and meet N in $I + 7V$. The following lemma shows that such a subgroup exists and is unique up to conjugation, and records its properties; the definition follows it.

Lemma 2.2. *Let $\bullet \in \{\text{ns}, \text{sp}\}$.*

- (1) *V is the direct sum of the scalar matrices and of the orthogonal complement of $\mathfrak{c}$ for the trace pairing $(A, B) \mapsto \text{tr}(AB)$. It has dimension 3, it is stable under conjugation by C^+ , and $V \cap \mathfrak{c}$ consists of the scalar matrices.*
- (2) *There is a subgroup G of $\text{GL}_2(\mathbb{Z}/49)$ that reduces onto C^+ and meets N in $I + 7V$, and any two such subgroups are conjugate by an element of N .*
- (3) *Such a group G has index 147 if $\bullet = \text{ns}$ and 196 if $\bullet = \text{sp}$, it has surjective determinant, and it contains $-I$.*

Proof. (1) The subalgebra $\mathfrak{c}_{\text{sp}}$ is spanned by $\begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$ and $\begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}$, and $\mathfrak{c}_{\text{ns}}$ by I and $J := \begin{pmatrix} 0 & \varepsilon \\ 1 & 0 \end{pmatrix}$. For $B = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$, the traces of these four products with B are a , d , $a + d$ and $b + \varepsilon c$. So the orthogonal complement of $\mathfrak{c}_{\text{sp}}$ is the space of matrices with zero diagonal, and that of $\mathfrak{c}_{\text{ns}}$ the space of matrices $\begin{pmatrix} a & -\varepsilon c \\ c & -a \end{pmatrix}$; neither contains a nonzero scalar, and adding the scalar matrices gives V_{sp} and (renaming c as $-b$) V_{ns} , which are thus the stated direct sums, of dimension 3. Conjugation by $g \in C^+$ fixes the scalars, preserves the trace pairing and normalizes C , hence $\mathfrak{c}$ and its orthogonal complement; so it preserves V . Finally, a diagonal matrix in V_{sp} is scalar, and $\begin{pmatrix} a & \varepsilon b \\ b & a \end{pmatrix} \in V_{\text{ns}}$ forces $b = -b = 0$.

(2) Modulo 49 we have $(I + 7A)(I + 7B) = I + 7(A + B)$ and $h(I + 7A)h^{-1} = I + 7\bar{h}A\bar{h}^{-1}$, with $\bar{h}$ the reduction of h . So $A \mapsto I + 7A$ is an isomorphism of $M_2(\mathbb{F}_7)$ onto N , and by (1) the subgroup $I + 7V$ is normal in the preimage P of C^+ in $\text{GL}_2(\mathbb{Z}/49)$. Put $Q := P/(I + 7V)$ and $\bar{N} := N/(I + 7V)$, a normal subgroup of Q of order 7 with $Q/\bar{N} \cong C^+$ of order 72 or 96, prime to 7. If G is as in the statement, then $I + 7V \subseteq G \subseteq P$, and $H := G/(I + 7V)$ is a complement of $\bar{N}$ in Q : $H \cap \bar{N} = (G \cap N)/(I + 7V)$ is trivial, and H maps onto $Q/\bar{N}$ because G reduces onto C^+ . Conversely, the preimage in P of a complement of $\bar{N}$ is a subgroup as in the statement. By the Schur–Zassenhaus theorem, complements of $\bar{N}$ exist, and any two are conjugate in $Q = \bar{N}H$, hence by an element of $\bar{N}$; taking preimages, any two subgroups as in the statement are conjugate by an element of N .

(3) As $G \cap N = I + 7V$ has order 7^3 and $G/(G \cap N) \cong C^+$, we have $\#G = 7^3 \#C^+$; since $\#\text{GL}_2(\mathbb{Z}/49) = 7^4 \#\text{GL}_2(\mathbb{F}_7) = 7^5 \cdot 288$, the index of G is $49 \cdot 288/72 = 196$ in the split case and $49 \cdot 288/96 = 147$ in the non-split case. Next, $\det(I + 7A) = 1 + 7\text{tr}(A)$ modulo 49 and $\text{tr}(aI) = 2a$, so $\det(G)$ contains $\det(I + 7V) = 1 + 7\mathbb{Z}/49$, the kernel of $(\mathbb{Z}/49)^\times \rightarrow \mathbb{F}_7^\times$; and $\det(G)$ reduces onto $\det(C^+) = \mathbb{F}_7^\times$, since $\det$ is ad on $C_{\text{sp}}(7)$ and the norm $\mathbb{F}_{49}^\times \rightarrow \mathbb{F}_7^\times$ on $C_{\text{ns}}(7)$. Hence $\det(G) = (\mathbb{Z}/49)^\times$. Finally, let $g \in G$ reduce to $-I$ and write $g = -(I + 7B)$. Then $g^2 = I + 14B \in G \cap N = I + 7V$, so $2B$, hence B , lies in V ; thus $I + 7B \in G$ and $-I = g(I + 7B)^{-1} \in G$. $\square$

Definition 2.3 ([FL26, Definition 2.7]). For $\bullet \in \{\text{ns}, \text{sp}\}$, let $G_\bullet^\sharp(49)$ be a subgroup of $\text{GL}_2(\mathbb{Z}/49)$ that reduces onto C^+ and meets the kernel of reduction in $I + 7V$; by Lemma 2.2 it is well defined up to conjugation, and so is the corresponding modular curve $X_\bullet^\sharp(49)$. These curves are $X_{\text{ns}}^\sharp(49) = \text{49.147.9.a.1}$ and $X_{\text{sp}}^\sharp(49) = \text{49.196.9.a.1}$ [FL26, Remark 2.8].

There are exactly seven such subgroups, all conjugate by N . In the proof of Lemma 2.2(2) they correspond to the complements H of $\bar{N}$ in Q . An element of $\bar{N}$ that normalizes H centralizes it, since $H \cap \bar{N}$ is trivial; but $\bar{N} \cong M_2(\mathbb{F}_7)/V \cong \mathfrak{c}/\mathbb{F}_7 I$ (as $\mathfrak{c}$ meets its orthogonal complement trivially), and the elements of C^+ outside C act on it by -1 . So the seven $\bar{N}$ -conjugates of H are distinct.

Lemma 2.4. *The nonzero nilpotent elements of V_{sp} are the nonzero multiples of $\begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$ and of $\begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}$. The space V_{ns} contains no nonzero nilpotent element.*

Proof. A matrix $A \in M_2(\mathbb{F}_7)$ is nilpotent if and only if $\text{tr}(A) = \det(A) = 0$. For $A = \begin{pmatrix} a & b \\ c & a \end{pmatrix} \in V_{\text{sp}}$ this says $2a = 0$ and $a^2 - bc = 0$, that is, $a = 0$ and $bc = 0$. For $A = \begin{pmatrix} a & \varepsilon b \\ -b & d \end{pmatrix} \in V_{\text{ns}}$ it says $d = -a$ and $-a^2 + \varepsilon b^2 = 0$. Since ε is not a square modulo 7, the second equation forces $b = 0$; hence $a = 0$ and $d = 0$. $\square$

2.3. Primes of potentially multiplicative reduction. The next proposition is the input from the arithmetic of elliptic curves. In the non-split case it is a special case of [FL26, Proposition 2.10], which generalizes [Fur24, Proposition 5.13]; the proof below follows theirs. In the split case we need to argue differently; see Remark 2.7. The following lemma isolates the modular input of the split case: where a point of $X_{\text{sp}}^+(7)$ with potentially multiplicative reduction reduces. Recall that $C_{\text{sp}}^+(7)$ is the stabilizer of the unordered pair of coordinate lines of $\mathbb{F}_7^2$ and contains $-I$. So a point of $X_{\text{sp}}^+(7)$ over a field k of characteristic 0, with $j \neq 0, 1728$, is given by a pair $(E, \{L_1, L_2\})$, where E is an elliptic curve over k and $\{L_1, L_2\}$ is a Gal_k -stable unordered pair of distinct lines in $E[7](\bar{k})$ (a [level structure](#) on E), up to isomorphism and quadratic twist [RSZB22, Section 2.3]. This is the moduli interpretation of the curve $X_{\text{split}}(7)$ of Mazur [Maz77], which is our $X_{\text{sp}}^+(7)$; similarly, Mazur's $X_{\text{non-split}}(7)$ is our $X_{\text{ns}}^+(7)$.

Lemma 2.5. *Let $p \neq 7$ be a prime, and let E be an elliptic curve over $\mathbb{Q}_p$ with potentially multiplicative reduction, so that E is a quadratic twist of a Tate curve E_q and $E[7](\overline{\mathbb{Q}_p}) = E_q[7](\overline{\mathbb{Q}_p})$ contains the line $\mu_7(\overline{\mathbb{Q}_p})$. Let $\{L_1, L_2\}$ be a $\text{Gal}_{\mathbb{Q}_p}$ -stable pair of distinct lines in $E[7](\overline{\mathbb{Q}_p})$, and let $t \in X_{\text{sp}}^+(7)(\mathbb{Q}_p)$ be the point defined by $(E, \{L_1, L_2\})$. If $\mu_7(\overline{\mathbb{Q}_p}) \in \{L_1, L_2\}$, then t reduces modulo p to the rational cusp $t = \infty$: writing $t = (x : y)$ with $x, y \in \mathbb{Z}_p$ not both in $p\mathbb{Z}_p$, we have $p \mid y$.*

Proof. We first describe the cusps. They correspond to the orbits of the group of matrices $\pm \begin{pmatrix} 1 & * \\ 0 & 1 \end{pmatrix}$ on the 28 unordered pairs $\{L_1, L_2\}$ of distinct lines in $\mathbb{F}_7^2$, in a basis whose first vector spans the subgroup scheme μ_7 of the Tate curve E_q over $\mathbb{Z}[1/7](\langle q \rangle)$ [DR73, VI, 5.1 and Construction 5.3, and VII, 1.13–1.16], [KM85, Corollary 10.2.5]. Indeed, the 7-torsion of the Tate curve has the basis $(\zeta_7, q^{1/7})$ over $\mathbb{Z}[1/7][\zeta_7](\langle q^{1/7} \rangle)$. The automorphism of this ring over $\mathbb{Z}[1/7][\zeta_7](\langle q \rangle)$ that fixes ζ_7 and sends $q^{1/7}$ to $\zeta_7 q^{1/7}$ acts on this basis by $\begin{pmatrix} a & 0 \\ 0 & 1 \end{pmatrix}$, which generates the unipotent matrices, and $-I$ comes from the automorphism -1 of E_q . The seven pairs containing the line μ_7 form one orbit, and the other 21 pairs form three. The group $\text{Gal}_{\mathbb{Q}}$ acts on the cusps through the cyclotomic character, by the matrices $\begin{pmatrix} a & 0 \\ 0 & 1 \end{pmatrix}$ with $a \in \mathbb{F}_7^\times$ [DR73, VI, 5.2–5.3]. These fix the first orbit and permute the other three transitively, so the first orbit is the unique rational cusp. In Zywin's coordinate $\mathbf{t}$ this cusp is $t = \infty$, and the other three are the roots of f_{sp} , the remaining poles of j_{sp} in (2.2).

Next, the model. Over $\mathbb{Z}[1/7]$ the coarse moduli scheme of $X_{\text{sp}}^+(7)$ is smooth [DR73, VI, Proposition 6.7], [KM85, Theorem 10.9.7], and proper, being finite over the j -line; its cusps form a finite étale subscheme [KM85, Theorem 10.9.4(3)]. Let $\mathcal{X}$ be its base change to $\mathbb{Z}_p$. By properness the rational cusp extends to a section of $\mathcal{X}$, and a smooth proper curve over $\mathbb{Z}_p$ with fibers of genus 0 and a section is isomorphic to $\mathbb{P}_{\mathbb{Z}_p}^1$. So $\mathcal{X} = \mathbb{P}_{\mathbb{Z}_p}^1$ in some coordinate $\mathbf{t}'$, and the cusps have pairwise distinct reductions in $\mathbf{t}'$. They also have pairwise distinct reductions in Zywin's coordinate $\mathbf{t}$: the roots of the monic cubic f_{sp} are p -integral, so they

do not reduce to ∞ , and their reductions are distinct because $\text{disc}(f_{\text{sp}}) = 7^2$. We show that two such coordinates differ by an element of $\text{PGL}_2(\mathbb{Z}_p)$. The rational cusp is a $\mathbb{Z}_p$ -point in $\mathfrak{t}'$, and $\text{PGL}_2(\mathbb{Z}_p)$ acts transitively on $\mathbb{P}^1(\mathbb{Z}_p)$, so after composing $\mathfrak{t}'$ with an element of $\text{PGL}_2(\mathbb{Z}_p)$ we may assume that the rational cusp is $\mathfrak{t}' = \infty$. Then $\mathfrak{t}' = at + b$ with $a \in \mathbb{Q}_p^\times$ and $b \in \mathbb{Q}_p$, and the other three cusps are $a\theta_i + b$, for the roots θ_i of f_{sp} . These are p -integral, since they do not reduce to ∞ , and their differences $a(\theta_i - \theta_j)$ are units, since their reductions are distinct. Since $\theta_i - \theta_j$ is a unit, so is a , and $b = (a\theta_1 + b) - a\theta_1$ is integral. Hence $\mathcal{X} = \mathbb{P}_{\mathbb{Z}_p}^1$ in the coordinate $\mathfrak{t}$ as well, and reduction modulo p of $X_{\text{sp}}^+(7)(\mathbb{Q}_p)$ is computed in $\mathfrak{t}$.

Finally, the reduction of t . Since $-I \in C_{\text{sp}}^+(7)$, the point t depends only on the pair of subgroups $\{L_1, L_2\}$ of $E[7]$ and on E up to quadratic twist; so we may assume that $E = E_q$, and twisting does not move the line μ_7 . Since $v_p(j(E)) < 0$, the point t reduces to a cusp $\mathfrak{c}$ of the special fiber of $\mathcal{X}$. The formal completion of $\mathcal{X}$ along $\mathfrak{c}$ is described by the Tate curve: the points of $X_{\text{sp}}^+(7)(\mathbb{Q}_p)$ that reduce to $\mathfrak{c}$ are the points $(E_{q'}, \{L'_1, L'_2\})$ with $v_p(q') > 0$ and $\{L'_1, L'_2\}$ in the orbit $\mathfrak{c}$, the lines being read in the basis $(\zeta_7, q'^{1/7})$ of $E_{q'}[7]$ [DR73, VII, Corollaire 2.2], [KM85, Theorem 8.11.10]. This representation of a point is unique up to the automorphism -1 : the point determines q' , since $j(E_{q'})$ is its j -invariant, and it determines the pair of lines up to $\text{Aut}(E_{q'}) = \{\pm 1\}$, since $j \neq 0, 1728$; and -1 fixes every line. Applied to $t = (E_q, \{L_1, L_2\})$, this says that $\{L_1, L_2\}$ lies in the orbit $\mathfrak{c}$; so $\mathfrak{c}$ is the rational cusp if and only if $\mu_7(\overline{\mathbb{Q}}_p) \in \{L_1, L_2\}$. $\square$

Proposition 2.6. *Let $\bullet \in \{\text{ns}, \text{sp}\}$, and let E be an elliptic curve over $\mathbb{Q}$ with a basis (P_1, P_2) of $E[49](\overline{\mathbb{Q}})$ with respect to which $\rho_{E,49}(\text{Gal}_{\mathbb{Q}}) \subseteq G_{\bullet}^{\#}(49)$. The basis $(7P_1, 7P_2)$ of $E[7](\overline{\mathbb{Q}})$, with respect to which $\rho_{E,7}(\text{Gal}_{\mathbb{Q}}) \subseteq C_{\bullet}^+(7)$, defines a rational point t of $X_{\bullet}^+(7)$ with $j_{\bullet}(t) = j(E)$. In the split case, t is the point $(E, \{L_1, L_2\})$, where $L_1, L_2 \subset E[7](\overline{\mathbb{Q}})$ are the lines spanned by $7P_1$ and $7P_2$; the pair $\{L_1, L_2\}$ is $\text{Gal}_{\mathbb{Q}}$ -stable. Write $t = (x : y)$ with x, y coprime integers. Then $v_p(F_{\bullet}(x, y)) \equiv 0 \pmod{7}$ for every prime $p \notin \{2, 7\}$. (In the split case this also holds for $p = 2$, since $F_{\text{sp}}(x, y)$ is odd by Lemma 2.1(3).)*

Proof. Fix such a prime p and suppose that

$$m := v_p(F_{\bullet}(x, y)) > 0.$$

We show that $7 \mid m$.

Step 1: E has potentially multiplicative reduction at p , with Tate parameter q satisfying $v_p(q) = 7m$. By Lemma 2.1(1), $p \nmid H_{\text{ns}}(x, y)$ in the non-split case, and $p \nmid xyH_{\text{sp}}(x, y)$ in the split case. The formulas $j_{\text{ns}} = H_{\text{ns}}^3/F_{\text{ns}}^7$ and $j_{\text{sp}} = xH_{\text{sp}}^3/(yF_{\text{sp}})^7$ then give

$$v_p(j(E)) = -7m < 0,$$

so E has potentially multiplicative reduction at p . By Tate's theory [Sil94, Chapter V, Lemma 5.1 and Theorem 5.3], there is a unique $q \in \mathbb{Q}_p^\times$ with $v_p(q) > 0$ and $j(E_q) = j(E)$, and a character $\chi: \text{Gal}_{\mathbb{Q}_p} \rightarrow \{\pm 1\}$ such that $E_{\mathbb{Q}_p}$ is the quadratic twist of the Tate curve E_q by χ . Since $j(E_q) = q^{-1} + 744 + 196884q + \dots$,

$$v_p(q) = -v_p(j(E)) = 7m.$$

Step 2: the image of inertia contains the matrix $(\begin{smallmatrix} 1 & 14m \\ 0 & 1 \end{smallmatrix})$, in a basis of $E[49](\overline{\mathbb{Q}}_p)$ whose first vector spans $\mu_{49}(\overline{\mathbb{Q}}_p)$. Tate's uniformization $E_q(\overline{\mathbb{Q}}_p) \cong \overline{\mathbb{Q}}_p^\times/q^{\mathbb{Z}}$ is $\text{Gal}_{\mathbb{Q}_p}$ -equivariant, so the classes of ζ_{49} and $q^{1/49}$ form a basis of $E_q[49](\overline{\mathbb{Q}}_p)$. Write $q = p^{7m}u$ with $u \in \mathbb{Z}_p^\times$, and let

$c: I_p \rightarrow \mathbb{Z}/49$ be the surjective character of the inertia group defined by $\tau(p^{1/49}) = \zeta_{49}^{c(\tau)} p^{1/49}$. Since $p \neq 7$, both ζ_{49} and $u^{1/49}$ lie in an unramified extension of $\mathbb{Q}_p$, so for $\tau \in I_p$, we have that $\tau(\zeta_{49}) = \zeta_{49}$ and $\tau(q^{1/49}) = \zeta_{49}^{7mc(\tau)} q^{1/49}$. That is,

$$\rho_{E,49}(\tau) = \chi(\tau) \begin{pmatrix} 1 & 7mc(\tau) \\ 0 & 1 \end{pmatrix}$$

in this basis. Choose τ with $c(\tau) = 1$. Since $\chi(\tau)^2 = 1$,

$$\rho_{E,49}(\tau^2) = \begin{pmatrix} 1 & 14m \\ 0 & 1 \end{pmatrix} = I + 14mN_0, \quad N_0 := \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}.$$

Step 3: $m\bar{N} \in V$ for a nonzero nilpotent $\bar{N} \in M_2(\mathbb{F}_7)$ whose image is the line $\mu_7(\overline{\mathbb{Q}}_p)$. Identify $E[49](\overline{\mathbb{Q}})$ with $E[49](\overline{\mathbb{Q}}_p)$ through a fixed embedding of $\overline{\mathbb{Q}}$ into $\overline{\mathbb{Q}}_p$, and let $B \in \mathrm{GL}_2(\mathbb{Z}/49)$ be the matrix of the change from the basis of Step 2 to the basis of the statement, in which $\rho_{E,49}(\mathrm{Gal}_{\overline{\mathbb{Q}}}) \subseteq G^\sharp$. In the latter basis

$$\rho_{E,49}(\tau^2) = B(I + 14mN_0)B^{-1} = I + 7 \cdot 2m\bar{N}, \quad \bar{N} := BN_0B^{-1} \bmod 7.$$

The matrix $\bar{N}$ is nonzero and nilpotent, and its image is spanned by the class of ζ_7 , i.e., it is the line $\mu_7(\overline{\mathbb{Q}}_p) \subset E[7](\overline{\mathbb{Q}}_p)$. Since $\rho_{E,49}(\tau^2)$ lies in the kernel of reduction of $G^\sharp$, which is $I + 7V$, we get $2m\bar{N} \in V$; hence $m\bar{N} \in V$.

Step 4: if $7 \nmid m$, then $\bullet = \mathrm{sp}$ and $\mu_7(\overline{\mathbb{Q}}_p)$ is one of the two lines of the level structure of $t = x/y$. If $7 \nmid m$, then $\bar{N} \in V$ is a nonzero nilpotent matrix. By Lemma 2.4 this is impossible when $\bullet = \mathrm{ns}$. When $\bullet = \mathrm{sp}$, the same lemma shows that $\bar{N}$ is a multiple of $\begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$ or of $\begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$, so its image $\mu_7(\overline{\mathbb{Q}}_p)$ is one of the two coordinate lines $L_1, L_2 \subset E[7](\overline{\mathbb{Q}})$, the two lines of the level structure defining $t = x/y$.

Step 5: in the situation of Step 4, $p \mid y$; this is a contradiction. By Step 4 and Lemma 2.5, applied to $E_{\mathbb{Q}_p}$ and the pair $\{L_1, L_2\}$, the point $t = x/y$ reduces modulo p to the rational cusp, that is, $p \mid y$. But $p \mid F_{\mathrm{sp}}(x, y)$ and $\gcd(y, F_{\mathrm{sp}}(x, y)) = 1$ by Lemma 2.1(1). Hence $7 \mid m$. $\square$

Remark 2.7. We explain why the split case is treated differently. The proof of [FL26, Proposition 2.10] is local: for an elliptic curve over a number field L and a prime $\mathfrak{p}$ of L of potentially multiplicative reduction, it first adjoins ζ_7 to L and then considers the image of the whole local Galois group of the completion $L(\zeta_7)_{\mathfrak{P}}$ at a prime $\mathfrak{P}$ above $\mathfrak{p}$. In the split case it uses an element τ of that group with $\chi_{49}(\tau) = 1 + 7$, where χ_{49} is the mod-49 cyclotomic character, whose existence is attributed there to the tameness of 7 in L . Such an element exists if and only if $\zeta_{49} \notin L(\zeta_7)_{\mathfrak{P}}$, a condition on $\mathfrak{p}$ rather than on the primes above 7. For $L = \mathbb{Q}$ it fails exactly at the primes p with $p^6 \equiv 1 \pmod{49}$, among them the primes $p \equiv \pm 1 \pmod{49}$, and such primes can divide $F_{\mathrm{sp}}(x, y)$. Proposition 2.6 recovers the congruence $v_p(F_{\mathrm{sp}}(x, y)) \equiv 0 \pmod{7}$ at every prime $p \neq 7$ using only inertia, together with the position of the rational cusp (Lemma 2.5). The same analysis shows that the rational cusp $t = \infty$ of $X_{\mathrm{sp}}^+(7)$ is *unramified* in $X_{\mathrm{sp}}^\sharp(49) \rightarrow X_{\mathrm{sp}}^+(7)$, while the other three cusps are totally ramified: the cusp widths of $X_{\mathrm{sp}}^\sharp(49)$ are 7 (seven times) and 49 (three times), and those of $X_{\mathrm{ns}}^\sharp(49)$ are 49 (three times). Accordingly, the further assertion of the split case of [FL26, Proposition 3.1], that y is a seventh power, is established by their argument only at the primes $p \mid y$ with $p^6 \not\equiv 1 \pmod{49}$. Neither [FL26] nor the present paper uses that assertion.

Proposition 2.8 ([FL26, Proposition 3.1]). *Let E be an elliptic curve over $\mathbb{Q}$ without complex multiplication.*

- (1) *If $\rho_{E,49}(\text{Gal}_{\mathbb{Q}})$ is conjugate to a subgroup of $G_{\text{ns}}^{\sharp}(49)$, there are coprime integers x, y and an integer w with $j(E) = j_{\text{ns}}(x/y)$ and $F_{\text{ns}}(x, y) = kw^7$, where $k \in \{7, 56\}$.*
- (2) *If $\rho_{E,49}(\text{Gal}_{\mathbb{Q}})$ is conjugate to a subgroup of $G_{\text{sp}}^{\sharp}(49)$, there are coprime integers x, y and an integer w with $j(E) = j_{\text{sp}}(x/y)$ and $F_{\text{sp}}(x, y) = kw^7$, where $k \in \{1, 7\}$.*

Proof. Let $t = x/y$ be the rational point of $X_{\bullet}^+(7)$ defined by E and its level structure, as in Proposition 2.6; then $j(E) = j_{\bullet}(t)$, because $j_{\bullet}$ is the j -map of $X_{\bullet}^+(7)$ in Zywna's coordinate [Zyw15, Section 1.4], [FL26, Theorem 2.1]. By Proposition 2.6 and Lemma 2.1(2)–(3), and since -1 is a seventh power, $F_{\bullet}(x, y) = kw^7$ with $k \in \{1, 7, 8, 56\}$ when $\bullet = \text{ns}$ and $k \in \{1, 7\}$ when $\bullet = \text{sp}$. In the non-split case, [FL26, Lemma 2.11] shows that $v_7(j(E)) > 0$. If $7 \nmid x$ then $7 \nmid F_{\text{ns}}(x, y)$ and $H_{\text{ns}}(x, y) \equiv 20x^7 \not\equiv 0 \pmod{7}$, so $v_7(j(E)) = 0$. Hence $7 \mid x$, and $7 \mid k$. $\square$

3. KUMMER DESCENT

We follow Bruin [Bru02, Section 3.1], [Bru06, Section 3]. Throughout this section we fix $\bullet \in \{\text{ns}, \text{sp}\}$, write $f = f_{\bullet}$, $F = F_{\bullet}$, $\theta = \theta_{\bullet}$, and fix one of the values of k allowed by Table 3. The table also records $\text{disc}(f)$ and the set $\mathcal{S}$ of [bad primes](#), those dividing $k \text{disc}(f)$. Its last two columns give generators of the group $\mathcal{O}_{K,\mathcal{S}}^{\times}$ of $\mathcal{S}$ -units of K , and the order of the Selmer group $K(7, \mathcal{S})$, which is defined below.

$\bullet$	f	$\text{disc}(f)$	k	$\mathcal{S}$	$\mathcal{O}_{K,\mathcal{S}}^{\times}$	$\#K(7, \mathcal{S})$
ns	$t^3 - 7t^2 + 7t + 7$	$2^6 7^2$	1, 7, 8, 56	$\{2, 7\}$	$\pm\langle \varepsilon_1, \varepsilon_2, \varpi, 2 \rangle$	7^4
sp	$t^3 - 4t^2 + 3t + 1$	7^2	1, 7	$\{7\}$	$\pm\langle \varepsilon_1, \varepsilon_2, \varpi \rangle$	7^3

TABLE 3. The data of the descent. In both rows $\varepsilon_1 := \theta_{\text{sp}} - 1$ and $\varepsilon_2 := 2 - \theta_{\text{sp}}$ are units, and $\varpi := 1 + \theta_{\text{sp}}$ has norm 7.

3.1. The covering curves. Since K has class number one, we may define the [\(7, \$\mathcal{S}\$ \)-Selmer group](#) of [Bru06, Section 3] as

$$K(7, \mathcal{S}) := \mathcal{O}_{K,\mathcal{S}}^{\times} / \mathcal{O}_{K,\mathcal{S}}^{\times 7},$$

of the order given in Table 3: by Dirichlet's unit theorem, $\mathcal{O}_{K,\mathcal{S}}^{\times}$ has rank $2 + \#\mathcal{S}$, since 2 is inert and 7 is totally ramified in K , and its torsion is ± 1 ; generators are listed in Table 3. We fix a set of representatives in $\mathcal{O}_{K,\mathcal{S}}^{\times}$ for $K(7, \mathcal{S})$, and we let $\Delta(\theta, k)$ be the subset of those representatives δ for which

$$\text{Norm}(\delta)/k \in \mathbb{Q}^{\times 7};$$

this condition only depends on the class of δ , since $\text{Norm}(\nu^7) \in \mathbb{Q}^{\times 7}$. The subset $\Delta(\theta, k)$ has 49 elements for each of the six pairs (F, k) . Let $\mathbf{a}_0, \mathbf{a}_1, \mathbf{a}_2$ be coordinates on $\mathbb{P}_{\mathbb{Q}}^2$, and put

$$\begin{aligned} \mathbf{u}_0 &:= \mathbf{a}_0 + \mathbf{a}_1\theta + \mathbf{a}_2\theta^2, \\ \mathbf{u}_1 &:= \mathbf{a}_0 + \mathbf{a}_1\sigma(\theta) + \mathbf{a}_2\sigma(\theta)^2, \\ \mathbf{u}_2 &:= \mathbf{a}_0 + \mathbf{a}_1\sigma^2(\theta) + \mathbf{a}_2\sigma^2(\theta)^2, \end{aligned}$$

in $K[\mathbf{a}_0, \mathbf{a}_1, \mathbf{a}_2]$, with the index i of $\mathbf{u}_i$ read modulo 3. Letting σ act on $K[\mathbf{a}_0, \mathbf{a}_1, \mathbf{a}_2]$ through the coefficients, we have $\sigma(\mathbf{u}_i) = \mathbf{u}_{i+1}$; and for $\omega = a_0 + a_1\theta + a_2\theta^2 \in K$ we have $\mathbf{u}_i(a_0, a_1, a_2) = \sigma^i(\omega)$. Thus $\mathbf{u}_0$ is the **generic element** of K , and $\mathbf{u}_0\mathbf{u}_1\mathbf{u}_2$ is its norm; we extend Tr and Norm to $K[\mathbf{a}_0, \mathbf{a}_1, \mathbf{a}_2]$ coefficientwise. For $\delta \in K^{\times}$ the **coordinate forms** $P_{\theta, \delta}^{(n)} \in \mathbb{Q}[\mathbf{a}_0, \mathbf{a}_1, \mathbf{a}_2]$ of degree 7 are defined by

$$(3.1) \quad \delta(\mathbf{a}_0 + \mathbf{a}_1\theta + \mathbf{a}_2\theta^2)^7 = P_{\theta, \delta}^{(0)} + P_{\theta, \delta}^{(1)}\theta + P_{\theta, \delta}^{(2)}\theta^2,$$

and we put

$$(3.2) \quad C_{\theta, \delta} := \{P_{\theta, \delta}^{(2)} = 0\} \subset \mathbb{P}_{\mathbb{Q}}^2.$$

Writing $\mathbf{a} = (a_0 : a_1 : a_2)$ for a point of $C_{\theta, \delta}$, we define

$$(3.3) \quad \phi_{\theta, \delta}: C_{\theta, \delta} \rightarrow \mathbb{P}_{\mathbb{Q}}^1, \quad \mathbf{a} \mapsto (P_{\theta, \delta}^{(0)}(\mathbf{a}) : -P_{\theta, \delta}^{(1)}(\mathbf{a})).$$

We identify $\mathbb{P}^2(\mathbb{Q})$ with $K^{\times}/\mathbb{Q}^{\times}$ through the bijection

$$(3.4) \quad \mathbb{P}^2(\mathbb{Q}) \rightarrow K^{\times}/\mathbb{Q}^{\times}, \quad \mathbf{a} = (a_0 : a_1 : a_2) \mapsto [a_0 + a_1\theta + a_2\theta^2].$$

Remark 3.1 (Intrinsic description). Write $\mathcal{V}_K$ for K regarded as a $\mathbb{Q}$ -vector space. For a finite-dimensional $\mathbb{Q}$ -vector space W , let $\mathbb{P}(W) := \text{Proj Sym}(W^{\vee})$ be the projective space of lines in W , so that $\mathbb{P}(W)(\mathbb{Q}) = (W \setminus \{0\})/\mathbb{Q}^{\times}$; in particular $\mathbb{P}(\mathcal{V}_K)(\mathbb{Q}) = K^{\times}/\mathbb{Q}^{\times}$. The coordinates $\mathbf{a}_0, \mathbf{a}_1, \mathbf{a}_2$ form the basis of $\mathcal{V}_K^{\vee}$ dual to $1, \theta, \theta^2$, so $\text{Sym}(\mathcal{V}_K^{\vee}) = \mathbb{Q}[\mathbf{a}_0, \mathbf{a}_1, \mathbf{a}_2]$, and the bijection (3.4) is the resulting isomorphism $\mathbb{P}_{\mathbb{Q}}^2 \cong \mathbb{P}(\mathcal{V}_K)$. In these terms the objects of the descent do not depend on coordinates:

$$C_{\theta, \delta} = \{[\omega] \in \mathbb{P}(\mathcal{V}_K) : \delta\omega^7 \in \mathbb{Q} + \mathbb{Q}\theta\}, \quad \phi_{\theta, \delta}([\omega]) = [\delta\omega^7] \in \mathbb{P}(\mathbb{Q} + \mathbb{Q}\theta),$$

and $\mathbb{P}(\mathbb{Q} + \mathbb{Q}\theta)$ is identified with $\mathbb{P}_{\mathbb{Q}}^1$ by $x - \theta y \mapsto (x : y)$ (Lemma 3.2 below). Since $\mathbb{Q} + \mathbb{Q}\theta_{\text{ns}} = \mathbb{Q} + \mathbb{Q}\theta_{\text{sp}}$, the curves are the same for both roots, and only the coordinate on the target changes; see Lemma 3.6. Likewise, in Section 4 the quotient map and its target will be $\psi([\omega]) = [\omega^3\sigma^{-1}(\omega)]$ and $Z_{\theta, \delta} = \{[v] \in \mathbb{P}(\mathcal{V}_K) : \text{Tr}(\eta v^3\sigma(v)) = 0\}$, with $\eta = \delta/f'(\theta)$. The coordinate forms $P_{\theta, \delta}^{(n)}$ of (3.1) and $Q_{\theta, \delta}^{(n)}$ of (4.3) are the equations of these objects in the basis $1, \theta, \theta^2$; we keep them because they are what the computations use.

Finally, let $\mathcal{U}$ be the complement of $\{x = y = 0\}$ in the hypersurface $\{F(x, y) = kw^7\} \subset \mathbb{A}_{\mathbb{Z}}^3$, so that

$$\mathcal{U}(\mathbb{Z}) = \{(x, y, w) \in \mathbb{Z}^3 : F(x, y) = kw^7 \text{ and } \gcd(x, y) = 1\},$$

and let

$$\pi: \mathcal{U}_{\mathbb{Q}} \rightarrow \mathbb{P}_{\mathbb{Q}}^1, \quad (x, y, w) \mapsto (x : y).$$

Lemma 3.2 (Choice of sign). *Let $\delta, \omega \in K^{\times}$ and $x, y \in \mathbb{Q}$ with $x - \theta y = \delta\omega^7$. Then $[\omega] \in C_{\theta, \delta}(\mathbb{Q})$ and $\phi_{\theta, \delta}([\omega]) = (x : y)$. Thus the common target of π and $\phi_{\theta, \delta}$ is the $\mathbb{P}^1$ with Zywin's coordinate $\mathbf{t} = x/y$.*

Proof. By (3.1), the coefficients of $1, \theta, \theta^2$ in $\delta\omega^7$ are the values at the coordinates of ω of $P_{\theta,\delta}^{(0)}$, $P_{\theta,\delta}^{(1)}$ and $P_{\theta,\delta}^{(2)}$. So $P_{\theta,\delta}^{(2)} = 0$, $P_{\theta,\delta}^{(0)} = x$ and $P_{\theta,\delta}^{(1)} = -y$, and $\phi_{\theta,\delta}([\omega]) = (x : y)$; when $(x, y, w) \in \mathcal{U}(\mathbb{Z})$ this is $\pi(x, y, w)$. $\square$

Proposition 3.3 ([Bru06, Theorem 3], [Bru02, Lemma 3.1.2]). *With notation as above,*

$$\pi(\mathcal{U}(\mathbb{Z})) \subseteq \bigcup_{\delta \in \Delta(\theta, k)} \phi_{\theta,\delta}(C_{\theta,\delta}(\mathbb{Q})).$$

More precisely, for $(x, y, w) \in \mathcal{U}(\mathbb{Z})$ there are $\delta \in \Delta(\theta, k)$ and $\omega \in K^\times$ with $x - \theta y = \delta\omega^7$; then $[\omega] \in C_{\theta,\delta}(\mathbb{Q})$ and $\phi_{\theta,\delta}([\omega]) = (x : y)$.

Proof. Let $(x, y, w) \in \mathcal{U}(\mathbb{Z})$. Let $\mathfrak{p}$ be a prime of $\mathcal{O}_K$ not above $\mathcal{S}$, and let p be the rational prime below it; p is unramified in K . If $\mathfrak{p}$ divides both $x - \theta y$ and a conjugate $x - \sigma^i(\theta)y$ with $i \neq 0$, then it divides $(\theta - \sigma^i(\theta))x$ and $(\theta - \sigma^i(\theta))y$, hence, since $\gcd(x, y) = 1$, the element $\theta - \sigma^i(\theta)$, which divides $\text{disc}(f)$; this is a contradiction. Applying powers of σ , we see that the three ideals $\sigma^i(x - \theta y)\mathcal{O}_{K,\mathcal{S}}$ are pairwise coprime. Their product is $F(x, y)\mathcal{O}_{K,\mathcal{S}} = kw^7\mathcal{O}_{K,\mathcal{S}} = w^7\mathcal{O}_{K,\mathcal{S}}$, since k is an $\mathcal{S}$ -unit, so each of them is a seventh power. Since K has class number one, $x - \theta y = \delta\omega^7$ with $\delta \in \mathcal{O}_{K,\mathcal{S}}^\times$, which we may take among the fixed representatives. Taking norms, $\text{Norm}(\delta)/k = (w/\text{Norm}(\omega))^7$, so $\delta \in \Delta(\theta, k)$. Finally, the coefficient of θ^2 in $\delta\omega^7 = x - \theta y$ is 0, so $[\omega] \in C_{\theta,\delta}(\mathbb{Q})$, and $\phi_{\theta,\delta}([\omega]) = (x : y)$ by Lemma 3.2. $\square$

3.2. The trace form. Since $K/\mathbb{Q}$ is Galois, a point $\omega \in K$ can be read through its three conjugates $\sigma^i(\omega) = \mathbf{u}_i(a_0, a_1, a_2)$. Accordingly, the linear forms $\mathbf{u}_i$ are coordinates on $\mathbb{P}^2$ over K , and Galois acts on them, through the coefficients, by cyclic permutation. Indeed, if $\tau \in \text{Gal}_{\mathbb{Q}}$ restricts to σ^a on K , then $\tau(\mathbf{u}_i) = \mathbf{u}_{i+a}$, indices modulo 3. In these coordinates the trace becomes a diagonal form, and the curves $C_{\theta,\delta}$ become twists of the Fermat septic

$$\mathcal{F}: \mathbf{u}_0^7 + \mathbf{u}_1^7 + \mathbf{u}_2^7 = 0.$$

Throughout, we put $\eta := \delta/f'(\theta)$, and we write $\phi_0: C_0 \rightarrow \mathbb{P}_{\mathbb{Q}}^1$ for $\phi_{\theta,f'(\theta)}$, the subscript 0 referring to the **trivial twist** $\eta = 1$; by Lemma 3.4 below, $C_0 = C_{\theta,f'(\theta)}$ is the curve $\{\text{Tr}(\omega^7) = 0\}$. Here and below, a curve written as $\{\text{Tr}(\eta\omega^7) = 0\}$ is described by the condition on its points $[\omega]$, and its equation is $\text{Tr}(\eta\mathbf{u}_0^7) = 0$. The multiplicative map $\omega \mapsto \omega^3\sigma^{-1}(\omega)$ commutes with σ , hence is defined over $\mathbb{Q}$; it sends $\{\text{Tr}(\eta\omega^7) = 0\}$ to a twist $\{\text{Tr}(\eta v^3\sigma(v)) = 0\}$ of the Klein quartic, and this is the subject of Section 4.

Lemma 3.4. *For $\delta \in K^\times$,*

$$P_{\theta,\delta}^{(2)} = \text{Tr}(\delta\mathbf{u}_0^7/f'(\theta)).$$

Proof. By Euler's formula [Neu99, Chapter III, Proposition 2.4], $\text{Tr}(\theta^i/f'(\theta))$ equals 0 for $i = 0, 1$ and 1 for $i = 2$. So $\text{Tr}(\cdot/f'(\theta))$ is the coefficient of θ^2 in the power basis. $\square$

The covering $\phi_{\theta,\delta}$ is geometrically Galois [Bru02, Theorem 3.1.1]: the group $\text{Aut}(\phi_{\theta,\delta,\overline{\mathbb{Q}}})$ of geometric automorphisms is $\mu_7(\overline{\mathbb{Q}})^3/\mu_7(\overline{\mathbb{Q}})$, acting by

$$(3.5) \quad (\mathbf{u}_0 : \mathbf{u}_1 : \mathbf{u}_2) \mapsto (\zeta_0\mathbf{u}_0 : \zeta_1\mathbf{u}_1 : \zeta_2\mathbf{u}_2).$$

Over $\mathbb{Q}$, the automorphism group scheme $\mathbf{Aut}(\phi_{\theta,\delta})$ is the twist $\text{Res}_{K/\mathbb{Q}}(\mu_7)/\mu_7$ of μ_7^3/μ_7 , acting by $[\omega] \mapsto [\zeta\omega]$.

Lemma 3.5. *Let $\delta \in K^\times$ and $\eta := \delta/f'(\theta)$.*

- (1) Over K , the forms u_i are linear coordinates on $\mathbb{P}^2$, in which $C_{\theta,\delta}$ becomes the diagonal septic

$$(3.6) \quad \eta u_0^7 + \sigma(\eta) u_1^7 + \sigma^2(\eta) u_2^7 = 0.$$

In particular $C_{\theta,\delta}$ is a twist of $\mathcal{F}$, a smooth plane curve of genus 15, and $\phi_{\theta,\delta}$ is the map of degree 49 of [Bru06, Theorem 3], branched over the three roots of f .

- (2) The covering $\phi_{\theta,\delta}$ is the twist of ϕ_0 by the image of $[\eta]$ under the isomorphism

$$K^\times / \mathbb{Q}^\times K^{\times 7} \cong H^1(\mathbb{Q}, \text{Aut}(\phi_{0,\overline{\mathbb{Q}}})),$$

and every twist of ϕ_0 arises in this way. In particular $(C_{\theta,\delta}, \phi_{\theta,\delta})$ depends, up to isomorphism, only on the class of η in $K^\times / \mathbb{Q}^\times K^{\times 7}$.

- (3) Over $\mathbb{Q}$, C_0 is the twist of $\mathcal{F}$ by the cocycle $\text{Gal}(K/\mathbb{Q}) \rightarrow \text{Aut}(\mathcal{F})$ sending σ to the cyclic permutation of the coordinates.

Proof. (1) By Lemma 3.4, $P_{\theta,\delta}^{(2)} = \text{Tr}(\eta u_0^7) = \sum_i \sigma^i(\eta u_0^7) = \sum_i \sigma^i(\eta) u_i^7$, which is (3.6). The forms are $u_i = \sum_n \sigma^i(\theta)^n a_n$, whose matrix $(\sigma^i(\theta)^n)_{i,n}$ is the Vandermonde matrix of the three roots of f . Since f is monic,

$$\det(\sigma^i(\theta)^n)_{i,n}^2 = \prod_{i < i'} (\sigma^i(\theta) - \sigma^{i'}(\theta))^2 = \text{disc}(f) \neq 0,$$

so $(a_0, a_1, a_2) \mapsto (u_0, u_1, u_2)$ is a linear change of coordinates over K . A diagonal form whose three coefficients are nonzero defines a smooth plane curve, which over $\overline{\mathbb{Q}}$ is isomorphic to $\mathcal{F}$ by rescaling the coordinates, and a smooth plane curve of degree 7 has genus $(7-1)(7-2)/2 = 15$. The last assertion is [Bru06, Theorem 3]; the degree also follows from the description of $\text{Aut}(\phi_{\theta,\delta,\overline{\mathbb{Q}}})$ in (3.5), since the covering is Galois.

(2) Identify $K \otimes_{\mathbb{Q}} \overline{\mathbb{Q}}$ with $\overline{\mathbb{Q}}^3$ via $a \otimes b \mapsto (\sigma^i(a)b)_i$, and choose $\xi = (\xi_0, \xi_1, \xi_2) \in (K \otimes_{\mathbb{Q}} \overline{\mathbb{Q}})^\times$ with $\xi^7 = \eta$, that is, $\xi_i^7 = \sigma^i(\eta)$. By (1), $(u_0 : u_1 : u_2) \mapsto (\xi_0 u_0 : \xi_1 u_1 : \xi_2 u_2)$ is an isomorphism $C_{\theta,\delta} \rightarrow C_0$ over $\overline{\mathbb{Q}}$. It commutes with the maps to $\mathbb{P}^1$, because it carries the conjugates $\sigma^i(\delta u_0^7) = \sigma^i(f'(\theta)) \sigma^i(\eta) u_i^7$, which determine $P_{\theta,\delta}^{(0)}$ and $P_{\theta,\delta}^{(1)}$, to those of $f'(\theta) u_0^7$. The associated cocycle $\tau \mapsto \tau(\xi)/\xi$ is the Kummer cocycle of η , with values in $\text{Res}_{K/\mathbb{Q}}(\mu_7)(\overline{\mathbb{Q}})$, pushed forward to $\text{Aut}(\phi_{0,\overline{\mathbb{Q}}}) = (\text{Res}_{K/\mathbb{Q}}(\mu_7)/\mu_7)(\overline{\mathbb{Q}})$. It remains to compute $H^1(\mathbb{Q}, \text{Aut}(\phi_{0,\overline{\mathbb{Q}}}))$. The exact sequence $1 \rightarrow \mu_7 \rightarrow \text{Res}_{K/\mathbb{Q}}(\mu_7) \rightarrow \text{Res}_{K/\mathbb{Q}}(\mu_7)/\mu_7 \rightarrow 1$, together with Shapiro's lemma and Hilbert 90, gives the exact sequence

$$\mathbb{Q}^\times / \mathbb{Q}^{\times 7} \rightarrow K^\times / K^{\times 7} \rightarrow H^1(\mathbb{Q}, \text{Aut}(\phi_{0,\overline{\mathbb{Q}}})) \rightarrow \text{Br}(\mathbb{Q})[7] \rightarrow \text{Br}(K)[7].$$

The last map is injective, since its composite with the corestriction is multiplication by $[K : \mathbb{Q}] = 3$. Hence $H^1(\mathbb{Q}, \text{Aut}(\phi_{0,\overline{\mathbb{Q}}})) \cong K^\times / \mathbb{Q}^\times K^{\times 7}$, and every class is the class of some η , that is, of $\phi_{\theta,\delta}$ with $\delta = \eta f'(\theta)$.

(3) For $\eta = 1$, (3.6) is the equation of $\mathcal{F}$, and $\tau \in \text{Gal}_{\mathbb{Q}}$ acts on the coordinates u_i through the cyclic permutation $u_i \mapsto u_{i+a}$, where $\tau|_K = \sigma^a$. $\square$

Lemma 3.6. *Let Σ be the image of $\mathcal{O}_K[1/7]^\times$ in $K^\times / \mathbb{Q}^\times K^{\times 7}$.*

- (1) *The pair $(C_{\theta,\delta}, \phi_{\theta,\delta})$ depends only on the class of $\eta = \delta/f'(\theta)$ in $K^\times / \mathbb{Q}^\times K^{\times 7}$: replacing δ by $r\nu^7\delta$ with $r \in \mathbb{Q}^\times$ and $\nu \in K^\times$ amounts to the change of variables $\omega \mapsto \nu\omega$.*

- (2) Σ has 49 elements, and for each of the six pairs (F, k) the map $\delta \mapsto [\delta/f'(\theta)]$ is a bijection $\Delta(\theta, k) \rightarrow \Sigma$.
- (3) The six descents involve the same family of plane septic $\{\text{Tr}(\eta\omega^7) = 0\}$, indexed by $[\eta] \in \Sigma$, and only the maps to $\mathbb{P}^1$ differ: in Zywinia's coordinates they are related by the Möbius transformation $t_{\text{ns}} = 5 - 2t_{\text{sp}}$.

Proof. (1) Since $r\nu^7\delta\omega^7 = r\delta(\nu\omega)^7$, the map $[\omega] \mapsto [\nu\omega]$ carries $C_{\theta, r\nu^7\delta}$ onto $C_{\theta, \delta}$ compatibly with the maps to $\mathbb{P}^1$, and the factor r changes neither the condition $P_{\theta, \delta}^{(2)} = 0$ nor the point $(P_{\theta, \delta}^{(0)} : -P_{\theta, \delta}^{(1)})$. (This is also Lemma 3.5(2).)

(2) Since 2 is inert in K , $\mathcal{O}_{K, \mathcal{S}}^\times = 2^\mathbb{Z} \times \mathcal{O}_K[1/7]^\times$ when $\mathcal{S} = \{2, 7\}$; and $f'(\theta) \in \mathbb{Q}^\times \mathcal{O}_K[1/7]^\times$ for both roots, by (1.5). So $\delta \mapsto [\delta/f'(\theta)]$ is a well-defined map $\Delta(\theta, k) \rightarrow \Sigma$. It is injective. Let $\delta, \delta' \in \Delta(\theta, k)$ with $\delta' = r\nu^7\delta$ for some $r \in \mathbb{Q}^\times$ and $\nu \in K^\times$. Taking norms gives $r^3 \in \mathbb{Q}^{\times 7}$, since $\text{Norm}(\delta)/k$ and $\text{Norm}(\delta')/k$ are seventh powers; hence $r \in \mathbb{Q}^{\times 7}$, so $\delta' \in \delta K^{\times 7}$. A seventh root in K of an $\mathcal{S}$ -unit is an $\mathcal{S}$ -unit, so $\delta' \in \delta \mathcal{O}_{K, \mathcal{S}}^{\times 7}$, and $\delta = \delta'$ because both are among the fixed representatives of $K(7, \mathcal{S})$. It remains to count. Let $\mathbb{Q}_\mathcal{S}^\times$ be the group of $\mathcal{S}$ -units of $\mathbb{Q}$. The norm induces a homomorphism $K(7, \mathcal{S}) \rightarrow \mathbb{Q}_\mathcal{S}^\times / \mathbb{Q}_\mathcal{S}^{\times 7}$, whose target has order $7^{\#\mathcal{S}}$, and $\Delta(\theta, k)$ is a set of representatives of the fiber above the class of k . This homomorphism is surjective, since $\text{Norm}(2) = 2^3$ and $\text{Norm}(1 + \theta_{\text{sp}}) = 7$; so every fiber has $\#K(7, \mathcal{S})/7^{\#\mathcal{S}} = 49$ elements, by Table 3. Finally, $\#\Sigma = 49$ by Lemma 5.2(2) below, whose proof is independent of this section. An injective map between two sets of 49 elements is a bijection.

(3) By (2), the set of classes $[\eta]$, $\delta \in \Delta(\theta, k)$, is Σ for all six pairs, and by (1) the curves are determined by these classes. In the intrinsic description of Remark 3.1, $C_{\theta, \delta} = \{[\omega] : \delta\omega^7 \in \mathbb{Q} + \mathbb{Q}\theta\}$, and $\mathbb{Q} + \mathbb{Q}\theta_{\text{ns}} = \mathbb{Q} + \mathbb{Q}\theta_{\text{sp}}$; so the curves are the same, and only the identification of $\mathbb{P}(\mathbb{Q} + \mathbb{Q}\theta)$ with $\mathbb{P}_t^1$ differs. Explicitly, $x - \theta_{\text{ns}}y = (x - 5y) - \theta_{\text{sp}}(-2y)$, so $(x : y)_{\text{ns}} = (x - 5y : -2y)_{\text{sp}}$, that is, $t_{\text{ns}} = 5 - 2t_{\text{sp}}$. $\square$

4. THE KLEIN QUARTIC QUOTIENT

Let $\mathcal{Z} : \mathbf{v}_0^3\mathbf{v}_1 + \mathbf{v}_1^3\mathbf{v}_2 + \mathbf{v}_2^3\mathbf{v}_0 = 0$ be the Klein quartic. Following Elkies [Elk99, Section 3.2, (3.5)–(3.6)], we consider

$$(4.1) \quad \psi : \mathcal{F} \rightarrow \mathcal{Z}, \quad (\mathbf{v}_0 : \mathbf{v}_1 : \mathbf{v}_2) := (\mathbf{u}_0^3\mathbf{u}_2 : \mathbf{u}_1^3\mathbf{u}_0 : \mathbf{u}_2^3\mathbf{u}_1),$$

which is well defined because of the identity

$$(4.2) \quad \mathbf{v}_0^3\mathbf{v}_1 + \mathbf{v}_1^3\mathbf{v}_2 + \mathbf{v}_2^3\mathbf{v}_0 = (\mathbf{u}_0\mathbf{u}_1\mathbf{u}_2)^3 (\mathbf{u}_0^7 + \mathbf{u}_1^7 + \mathbf{u}_2^7).$$

The map ψ has two properties that we use.

- (1) It is the quotient of $\mathcal{F}$ by the group H of order 7 generated by the geometric automorphism $(\mathbf{u}_0 : \mathbf{u}_1 : \mathbf{u}_2) \mapsto (\zeta_7\mathbf{u}_0 : \zeta_7^2\mathbf{u}_1 : \zeta_7^4\mathbf{u}_2)$. Indeed, $\mathbf{u}_0^3\mathbf{u}_2$, $\mathbf{u}_1^3\mathbf{u}_0$, $\mathbf{u}_2^3\mathbf{u}_1$ are the only monomials of degree 4 invariant under this automorphism, so ψ is constant on the orbits of H and factors through $\mathcal{F}/H$; and ψ has degree 7 = $\#H$ [Elk99, Section 3.2, p. 74], so the induced map $\mathcal{F}/H \rightarrow \mathcal{Z}$ is birational, hence an isomorphism, $\mathcal{Z}$ being smooth.
- (2) It commutes with the cyclic permutations of the coordinates $(\mathbf{u}_0, \mathbf{u}_1, \mathbf{u}_2) \mapsto (\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_0)$ and $(\mathbf{v}_0, \mathbf{v}_1, \mathbf{v}_2) \mapsto (\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_0)$.

The second property is what allows ψ to be twisted. By Lemma 3.5, over K and in the coordinates u_i , the curve $C_{\theta,\delta}$ is a diagonal septic on which σ acts by cyclic permutation of the coordinates. So ψ should induce a map from $C_{\theta,\delta}$ to a twist of $\mathcal{Z}$, defined over $\mathbb{Q}$. We now make this precise.

4.1. Twisting the quotient map. Take the coordinates u_i of Lemma 3.5 on the source, and write v_i for the same linear forms, regarded as coordinates on the target. The formula for ψ reads $v_i = u_i^3 u_{i-1} = \sigma^i(u_0^3 u_2)$, where $u_0^3 u_2 = u_0^3 \sigma^{-1}(u_0)$. This expression is Galois equivariant, so we obtain a rational map defined over $\mathbb{Q}$, which we denote by the same letter,

$$\psi: \mathbb{P}_{\mathbb{Q}}^2 \dashrightarrow \mathbb{P}_{\mathbb{Q}}^2, \quad [\omega] \mapsto [\omega^3 \sigma^{-1}(\omega)];$$

it is given by three quartic forms with rational coefficients, and over K it restricts to the map $\psi: \mathcal{F} \rightarrow \mathcal{Z}$ of (4.1) on $C_0 \times_{\mathbb{Q}} K \cong \mathcal{F}$. For $\delta \in K^\times$, define quartic forms $Q_{\theta,\delta}^{(n)} \in \mathbb{Q}[a_0, a_1, a_2]$ by

$$(4.3) \quad \delta u_0^3 u_1 = Q_{\theta,\delta}^{(0)} + Q_{\theta,\delta}^{(1)} \theta + Q_{\theta,\delta}^{(2)} \theta^2,$$

exactly as the forms $P_{\theta,\delta}^{(n)}$ were defined from u_0^7 in (3.1), and put

$$(4.4) \quad Z_{\theta,\delta} := \{Q_{\theta,\delta}^{(2)} = 0\} \subset \mathbb{P}_{\mathbb{Q}}^2, \quad \varphi_{\theta,\delta}: Z_{\theta,\delta} \dashrightarrow \mathbb{P}_{\mathbb{Q}}^1, \quad [v] \mapsto (Q_{\theta,\delta}^{(0)} : -Q_{\theta,\delta}^{(1)}).$$

As in Lemma 3.4, $Q_{\theta,\delta}^{(2)} = \text{Tr}(\eta u_0^3 u_1)$ with $\eta = \delta/f'(\theta)$; so $Z_{\theta,\delta} = \{\text{Tr}(\eta v^3 \sigma(v)) = 0\}$ in the notation of Section 3, and over K , in the coordinates v_i , it becomes

$$(4.5) \quad \eta v_0^3 v_1 + \sigma(\eta) v_1^3 v_2 + \sigma^2(\eta) v_2^3 v_0 = 0,$$

a twist of the Klein quartic $\mathcal{Z}$.

Lemma 4.1. *For every $\delta \in K^\times$ and $n \in \{0, 1, 2\}$,*

$$Q_{\theta,\delta}^{(n)} \circ \psi = (u_0 u_1 u_2)^3 P_{\theta,\delta}^{(n)}.$$

Consequently ψ maps $C_{\theta,\delta}$ to $Z_{\theta,\delta}$, and the diagram

$$\begin{array}{ccc} \mathcal{F} & \xrightarrow{\psi} & \mathcal{Z} \\ \cong \uparrow & & \uparrow \cong \\ C_{\theta,\delta} & \xrightarrow{\psi} & Z_{\theta,\delta} \\ & \searrow \phi_{\theta,\delta} & \downarrow \varphi_{\theta,\delta} \\ & & \mathbb{P}_{\mathbb{Q}}^1 \end{array}$$

commutes: the triangle over $\mathbb{Q}$, that is, $\phi_{\theta,\delta} = \varphi_{\theta,\delta} \circ \psi$, and the square over $\overline{\mathbb{Q}}$, where the vertical isomorphisms are $u_i \mapsto \xi_i u_i$ and $v_i \mapsto \xi_i^3 \xi_{i-1} v_i$ (indices modulo 3) in the coordinates of (3.6) and (4.5), for any $\xi_i \in \overline{\mathbb{Q}}$ with $\xi_i^7 = \sigma^i(\eta)$. On rational points, ψ and $\varphi_{\theta,\delta}$ are everywhere defined.

Proof. Substituting ψ replaces u_i by $u_i^3 u_{i-1}$, so it sends $u_0^3 u_1$ to $(u_0^3 u_2)^3 u_1^3 u_0 = (u_0 u_1 u_2)^3 u_0^7$; on points, $v = \omega^3 \sigma^{-1}(\omega)$ satisfies $v^3 \sigma(v) = \text{Norm}(\omega)^3 \omega^7$. In the group ring $\mathbb{Z}[\text{Gal}(K/\mathbb{Q})]$, this is the identity $(3 + \sigma)(3 + \sigma^{-1}) = 7 + 3(1 + \sigma + \sigma^2)$. Multiply by δ and compare (3.1)

with (4.3); this gives the displayed identity, hence the triangle. For the square, put $u'_i = \xi_i u_i$ and $v'_i = u_i'^3 u_{i-1}' = \xi_i^3 \xi_{i-1} v_i$. Then $\sum_i u_i'^7 = \sum_i \sigma^i(\eta) u_i^7$ and

$$\sum_i v_i'^3 v_{i+1}' = \sum_i \xi_i^{10} \xi_{i-1}^3 \xi_{i+1}^3 v_i^3 v_{i+1} = (\xi_0 \xi_1 \xi_2)^3 \sum_i \sigma^i(\eta) v_i^3 v_{i+1},$$

so the vertical maps are isomorphisms onto $\mathcal{F}$ and $\mathcal{Z}$, and they intertwine the two maps ψ . A rational point is the class of some $\omega \in K^\times$, and then $v = \omega^3 \sigma^{-1}(\omega) \neq 0$ and $\delta v^3 \sigma(v) \neq 0$. $\square$

Remark 4.2. The map $\psi: \mathcal{F} \rightarrow \mathcal{Z}$ is a cyclic unramified cover of degree 7; indeed $2 \cdot 15 - 2 = 7(2 \cdot 3 - 2)$. Its deck group, generated by $(u_0 : u_1 : u_2) \mapsto (\zeta_7 u_0 : \zeta_7^2 u_1 : \zeta_7^4 u_2)$, is one of the two subgroups of order 7 of $\text{Aut}(\phi_{\theta, \delta, \overline{\mathbb{Q}}}) = \mu_7(\overline{\mathbb{Q}})^3 / \mu_7(\overline{\mathbb{Q}})$ that are stable under the cyclic permutation of the coordinates, which is why ψ descends to $\mathbb{Q}$. The other one, generated by $(u_0 : u_1 : u_2) \mapsto (\zeta_7 u_0 : \zeta_7^4 u_1 : \zeta_7^2 u_2)$, has the invariant monomials $u_0^3 u_1, u_1^3 u_2, u_2^3 u_0$; it gives $[\omega] \mapsto [\omega^3 \sigma(\omega)]$ and a second twist of the Klein quartic, $\{\text{Tr}(\eta v^3 \sigma^{-1}(v)) = 0\}$. For $\eta = 1$ this second curve is the twist of $\mathcal{Z}$ by the cocycle $\text{Gal}(K/\mathbb{Q}) \rightarrow \text{Aut}(\mathcal{Z})$ sending σ to the inverse of the cyclic permutation, and it is not trivial: it has no points over $\mathbb{Q}_2$, whereas $Z_0 \cong \mathcal{Z}$ by Corollary 4.7. The choice of σ is therefore not a convention. Twists of the Klein quartic failing the Hasse principle are studied in [LGV24]; see [FLGS18] for the classification of the twists. The other six subgroups of order 7 of $\text{Aut}(\phi_{\theta, \delta, \overline{\mathbb{Q}}})$ form two orbits of three under the cyclic permutation of the coordinates, so the corresponding quotient maps of $C_{\theta, \delta}$ are defined over K but not over $\mathbb{Q}$. One orbit consists of the three subgroups generated by $(u_0 : u_1 : u_2) \mapsto (\zeta_7 u_0 : u_1 : u_2)$ and its permutations, whose quotient maps are the projections $(u_0 : u_1 : u_2) \mapsto (u_1 : u_2)$ onto $\mathbb{P}^1$; the other three subgroups act freely, and their quotients are hyperelliptic curves of genus 3. The coverings ψ are twisted forms of the covering of the Klein quartic attached to the isogeny $1 - \zeta_7$ of its Jacobian, as in the descent theory for cyclic covers of Poonen and Schaefer [PS97]; compare [PSS07, Section 10.1].

Lemma 4.3. *Let $\delta, \delta' \in K^\times$ with $\delta' = r \nu^3 \sigma(\nu) \delta$ for some $r \in \mathbb{Q}^\times$ and $\nu \in K^\times$. Then $[v] \mapsto [\nu v]$ is an isomorphism $Z_{\theta, \delta'} \rightarrow Z_{\theta, \delta}$ over $\mathbb{Q}$, and it commutes with the maps φ . Moreover, for every $\lambda \in K^\times$,*

$$(4.6) \quad \lambda^7 = \text{Norm}(\lambda) \cdot \nu^3 \sigma(\nu), \quad \nu = \lambda^2 / \sigma(\lambda),$$

so that $Z_{\theta, \delta}$ and $\varphi_{\theta, \delta}$ only depend on the class of $\delta / f'(\theta)$ in the quotient $K^\times / \mathbb{Q}^\times (K^\times)^{3+\sigma}$ of $K^\times / \mathbb{Q}^\times K^{\times 7}$, where $(K^\times)^{3+\sigma} := \{\nu^3 \sigma(\nu) : \nu \in K^\times\}$.

Proof. The first statement follows from $\delta' \nu^3 \sigma(v) = r \delta (\nu v)^3 \sigma(\nu v)$. For the second, $\nu^3 \sigma(\nu) = \lambda^6 \sigma(\lambda)^{-3} \sigma(\lambda)^2 \sigma^2(\lambda)^{-1} = \lambda^7 / \text{Norm}(\lambda)$. $\square$

4.2. The untwisted quartic Z_0 . We use the following identity of Klein [Kle78, §5]; see also [Elk99, (1.3)].

Proposition 4.4 (Klein's identity). *Let $\sqrt{-7} := \zeta_7 + \zeta_7^2 + \zeta_7^4 - \zeta_7^3 - \zeta_7^5 - \zeta_7^6$, and let*

$$(4.7) \quad \alpha := \sqrt{-7} (\zeta_7^4 - \zeta_7^{-4}) = 4c_1 + 2c_2 + c_4 = \theta_{\text{sp}}^2 - 5\theta_{\text{sp}} + 1 \in \mathcal{O}_K, \quad c_a := \zeta_7^a + \zeta_7^{-a}.$$

For $\vartheta = \frac{1}{7}(\nu_0 \alpha + \nu_1 \sigma(\alpha) + \nu_2 \sigma^2(\alpha)) \in K[\nu_0, \nu_1, \nu_2]$, with σ acting on the coefficients, we have

$$\text{Tr}(\vartheta^3 \sigma(\vartheta)) = \nu_0^3 \nu_1 + \nu_1^3 \nu_2 + \nu_2^3 \nu_0.$$

Proof. We have $\sigma(\alpha) = \sqrt{-7}(\zeta_7 - \zeta_7^{-1})$ and $\sigma^2(\alpha) = \sqrt{-7}(\zeta_7^2 - \zeta_7^{-2})$. Since $\sqrt{-7}/7 = -1/\sqrt{-7}$, the numbers $\alpha_i := \sigma^i(\alpha)/7$ are the entries $-(\zeta_7^a - \zeta_7^{-a})/\sqrt{-7}$, $a \in \{4, 1, 2\}$, of the symmetric matrix of order two in Klein's group of 168 automorphisms of his quartic.

Let $\Phi(\mathbf{v}_0, \mathbf{v}_1, \mathbf{v}_2) = \mathbf{v}_0^3 \mathbf{v}_1 + \mathbf{v}_1^3 \mathbf{v}_2 + \mathbf{v}_2^3 \mathbf{v}_0$. Klein's matrix

$$M := \begin{pmatrix} \alpha_1 & \alpha_2 & \alpha_0 \\ \alpha_2 & \alpha_0 & \alpha_1 \\ \alpha_0 & \alpha_1 & \alpha_2 \end{pmatrix} \in \mathrm{SL}_3(K)$$

satisfies $M^2 = I$ and $\Phi \circ M = \Phi$. Each row of M is the image under σ of the previous one. So, for the column vector $\mathbf{s} = (s_1, s_2, s_3) := (\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_0)$, whose entries are fixed by σ , the entries of $M\mathbf{s}$ are $\vartheta, \sigma(\vartheta), \sigma^2(\vartheta)$, since $\alpha_1 s_1 + \alpha_2 s_2 + \alpha_0 s_3 = \vartheta$. Hence $\Phi(\mathbf{s}) = \Phi(M\mathbf{s}) = \mathrm{Tr}(\vartheta^3 \sigma(\vartheta))$, and $\Phi(\mathbf{s}) = \Phi(\mathbf{v}_0, \mathbf{v}_1, \mathbf{v}_2)$ because Φ is invariant under cyclic permutations. (The identity can also be checked by expanding both sides.) $\square$

Lemma 4.5 (Hurwitz). *The rational points of the Klein quartic are the trivial ones:*

$$\mathcal{Z}(\mathbb{Q}) = \{(1 : 0 : 0), (0 : 1 : 0), (0 : 0 : 1)\}.$$

Proof. We summarize the argument of Hurwitz [Hur08], which reduces the statement to Fermat's Last Theorem for exponent 7, proved by Lamé [Lam40]. Let $(x : y : z) \in \mathcal{Z}(\mathbb{Q})$ with $x, y, z \in \mathbb{Z}$ coprime. If one coordinate vanishes, say $z = 0$, then $x^3 y = 0$, and the point is one of the three above. Suppose that $xyz \neq 0$, and let p be a prime. The three terms $x^3 y, y^3 z, z^3 x$ sum to zero, so the smallest of their p -adic valuations is attained at least twice. If p divided exactly one of x, y, z , say x , then $y^3 z$ would be the only term prime to p ; so every prime dividing xyz divides exactly two of x, y, z . If $p \mid x$ and $p \mid y$, then $v_p(z^3 x) = v_p(x) < v_p(x^3 y)$ forces $v_p(x) = 3v_p(y)$, and cyclically. Hence $|x| = a^3 c, |y| = b^3 a$, and $|z| = c^3 b$ for

$$a = \prod_{p \mid \gcd(x, y)} p^{v_p(y)}, \quad b = \prod_{p \mid \gcd(y, z)} p^{v_p(z)}, \quad c = \prod_{p \mid \gcd(z, x)} p^{v_p(x)}.$$

Replacing (x, y, z) by $(-x, -y, -z)$ if necessary, we may assume that $xyz > 0$, and then signs can be given to a, b, c so that $(x, y, z) = (a^3 c, b^3 a, c^3 b)$. Now (4.2) gives $(abc)^3(a^7 + b^7 + c^7) = 0$, with $abc \neq 0$, which contradicts Lamé's theorem. $\square$

Remark 4.6. In the language of this section, the proof lifts a rational point of $\mathcal{Z}$ to a rational point of $\mathcal{F}$ along the cyclic unramified cover $(u_0 : u_1 : u_2) \mapsto (u_0^3 u_2 : u_1^3 u_0 : u_2^3 u_1)$. Elkies [Elk99, Section 3.1] gives a proof that does not use Lamé's theorem: the quotient of $\mathcal{Z}$ by the cyclic permutation of the coordinates is an elliptic curve of conductor 49 whose Mordell–Weil group has order 2, as a 2-descent in the style of Fermat shows, and the fibers above its two rational points contain no rational points other than the three points of Lemma 4.5. Read in the opposite direction, through the same cover, this gives a proof of Lamé's theorem [Elk99, Section 3.2].

Corollary 4.7. *The [untwisted quartic](#) $Z_0 := \{\mathrm{Tr}(v^3 \sigma(v)) = 0\} = Z_{\theta, f'(\theta)}$ is isomorphic over $\mathbb{Q}$ to the Klein quartic, and*

$$Z_0(\mathbb{Q}) = \{[\alpha], [\sigma(\alpha)], [\sigma^2(\alpha)]\}.$$

Proof. By Proposition 4.4, $Z_0 \cong \mathcal{Z}$, with the three points above corresponding to $(1 : 0 : 0), (0 : 1 : 0), (0 : 0 : 1)$. These are the only rational points of $\mathcal{Z}$, by Lemma 4.5. $\square$

5. PROOF OF THEOREM 1

In this section (with the exception of Proposition 5.3) $\theta = \theta_{\text{sp}}$ and $f = f_{\text{sp}}$, and $\varpi = 1 + \theta$ is as in Table 3. Then

$$(5.1) \quad \text{Norm}(\varpi) = 7, \quad 7 = \varpi^3 \theta^{-2}, \quad \sigma(\varpi) = \varpi(4 - \varpi), \quad f'(\theta) = \varpi^2 \left(3 - \frac{2\varpi}{\theta} \right),$$

and ϖ is a uniformizer of the completion K_7 of K at its prime above 7, which is a totally ramified cubic extension of $\mathbb{Q}_7$ with residue field $\mathbb{F}_7$. The element $\theta = \varpi - 1$ is a unit.

5.1. A homomorphism. Recall from Lemma 3.6 the group Σ , the image of $\mathcal{O}_K[1/7]^\times$ in $K^\times/\mathbb{Q}^\times K^{\times 7}$. By Lemma 4.3, the quartic $Z_{\theta,\delta}$ is isomorphic to Z_0 over $\mathbb{Q}$, compatibly with the maps to $\mathbb{P}^1$, as soon as the class of $\eta = \delta/f'(\theta)$ lies in $(3 + \sigma)\Sigma$. We shall see in Lemma 5.2 that $\Sigma \cong \mathbb{F}_7^2$ is the sum of the eigenspaces of σ for the eigenvalues 2 and 4, and that $(3 + \sigma)\Sigma$ is the 2-eigenspace, because $3 + 4 = 0$ in $\mathbb{F}_7$. What we need is therefore a homomorphism $\Sigma \rightarrow \mathbb{F}_7$ that kills the 2-eigenspace and not the 4-eigenspace, and whose value on $(x - \theta y)/f'(\theta)$ we can compute. A nonzero homomorphism ϕ with $\phi \circ \sigma = 4\phi$ has the first property, since $2\phi(v) = \phi(\sigma v) = 4\phi(v)$ forces $\phi(v) = 0$ on the 2-eigenspace. The following 7-adic map κ has both properties.

Every $g \in K_7^\times$ can be written as $g = \varpi^a u$ with $a \in \mathbb{Z}$ and $u \equiv u_0(1 + b\varpi) \pmod{\varpi^2}$, where $u_0 \in \mathbb{F}_7^\times$ and $b \in \mathbb{F}_7$. We define

$$\kappa: K_7^\times \rightarrow \mathbb{F}_7, \quad \kappa(g) := b + 4a.$$

Lemma 5.1. *The map κ is a homomorphism. It is trivial on $\mathbb{Q}_7^\times$ and on seventh powers, and $\kappa \circ \sigma = 4\kappa$.*

Proof. Both a and b are additive, and seventh powers die in $\mathbb{F}_7$. A unit of $\mathbb{Z}_7$ is congruent to an element of $\mathbb{F}_7^\times$ modulo 7, hence modulo ϖ^3 , so $\kappa(\mathbb{Z}_7^\times) = 0$. By (5.1), $7 = \varpi^3 \theta^{-2}$ with $\theta^{-2} = (1 - \varpi)^{-2} \equiv 1 + 2\varpi \pmod{\varpi^2}$, so $\kappa(7) = 2 + 12 = 0$. Finally, σ acts trivially on the residue field, and $\sigma(\varpi) = \varpi(4 - \varpi) \equiv 4\varpi(1 - 2\varpi) \pmod{\varpi^3}$, since $4^{-1} = 2$ in $\mathbb{F}_7$. Hence, for $g = \varpi^a u$ as above, $\sigma(g) \equiv \varpi^a \cdot 4^a (1 - 2\varpi)^a \sigma(u)$ modulo ϖ^{a+2} , with $\sigma(u) \equiv u_0(1 + 4b\varpi) \pmod{\varpi^2}$, and $\kappa(\sigma(g)) = (4b - 2a) + 4a = 4b + 2a = 4\kappa(g)$. $\square$

The group Σ is an $\mathbb{F}_7[\sigma]$ -module, and $1 + \sigma + \sigma^2$ acts as zero on it, because norms are rational. As $t^2 + t + 1 = (t - 2)(t - 4)$ in $\mathbb{F}_7[t]$, the action of σ on Σ is diagonalizable, with eigenvalues among the primitive cube roots of unity 2 and 4 of $\mathbb{F}_7$; Lemma 5.2 shows that both occur.

Lemma 5.2. *Let $\varepsilon_1 = \theta - 1$ and $\varepsilon_2 = 2 - \theta$ be the units of Table 3. Then the following hold.*

- (1) $\mathcal{O}_K^\times = \pm \langle \varepsilon_1, \varepsilon_2 \rangle$, with $\sigma(\varepsilon_1) = \varepsilon_2^{-1}$ and $\sigma(\varepsilon_2) = -\varepsilon_1 \varepsilon_2^{-1}$.
- (2) Σ is generated by the classes of ε_1 and ε_2 , and $\mathcal{O}_K^\times \rightarrow \Sigma$ induces an isomorphism $\mathcal{O}_K^\times / \pm \mathcal{O}_K^{\times 7} \cong \Sigma$; in particular $\Sigma \cong \mathbb{F}_7^2$.
- (3) $\Sigma = \Sigma_2 \oplus \Sigma_4$, where Σ_2 is generated by $[\varepsilon_1 \varepsilon_2^2]$ and σ acts on it as multiplication by 2, and Σ_4 is generated by $[\varepsilon_1 \varepsilon_2^4]$ and σ acts on it as multiplication by 4.
- (4) κ induces a homomorphism $\Sigma \rightarrow \mathbb{F}_7$, with $\kappa(\varepsilon_1) = 3$ and $\kappa(\varepsilon_2) = 2$, and

$$\ker(\kappa|_\Sigma) = \Sigma_2 = (3 + \sigma)\Sigma.$$

Proof. In part (1), the two relations are a direct computation, and that -1 , ε_1 and ε_2 generate $\mathcal{O}_K^\times$ is checked with Magma. Since $\varpi^3 = 7\theta^2$, the class of ϖ in $K^\times/\mathbb{Q}^\times K^{\times 7}$ equals

that of θ^{-4} , and θ is a unit; so Σ is generated by the classes of units, hence by those of ε_1 and ε_2 . If a unit u lies in $\mathbb{Q}^\times K^{\times 7}$, say $u = r\nu^7$ with $r \in \mathbb{Q}^\times$ and $\nu \in K^\times$, then $\pm 1 = \text{Norm}(u) = r^3 \text{Norm}(\nu)^7$, so $r \in \pm \mathbb{Q}^{\times 7}$, and $u = \pm (s\nu)^7$ with $s \in \mathbb{Q}^\times$, where $s\nu$ is a unit because its seventh power $\pm u$ is a unit. Hence the kernel of $\mathcal{O}_K^\times \rightarrow \Sigma$ is $\pm \mathcal{O}_K^{\times 7}$, and (2) follows from (1). For (3), in additive notation, $\sigma[\varepsilon_1] = -[\varepsilon_2]$ and $\sigma[\varepsilon_2] = [\varepsilon_1] - [\varepsilon_2]$, so $\sigma[\varepsilon_1 \varepsilon_2^2] = 2[\varepsilon_1 \varepsilon_2^2]$ and $\sigma[\varepsilon_1 \varepsilon_2^4] = 4[\varepsilon_1 \varepsilon_2^4]$, and these two classes span $\Sigma \cong \mathbb{F}_7^2$. By Lemma 5.1, κ is well defined on Σ . From $\varepsilon_1 = \varpi - 2 \equiv -2(1 - 4\varpi)$ and $\varepsilon_2 = 3 - \varpi \equiv 3(1 - 5\varpi)$ modulo ϖ^2 we get $\kappa(\varepsilon_1) = 3$ and $\kappa(\varepsilon_2) = 2$, so $\kappa(\varepsilon_1 \varepsilon_2^2) = 0$ and $\kappa(\varepsilon_1 \varepsilon_2^4) = 4$. Finally $3 + \sigma$ acts as 5 on Σ_2 and as 0 on Σ_4 . $\square$

5.2. The 7-adic step.

Proposition 5.3. *Let $\bullet \in \{\text{ns}, \text{sp}\}$, and write $f = f_\bullet$, $F = F_\bullet$ and $\theta = \theta_\bullet$. Let x, y be coprime integers with $7 \mid F(x, y)$.*

(1) *We have*

$$\kappa\left(\frac{x - \theta y}{f'(\theta)}\right) = 0.$$

(2) *If moreover $v_p(F(x, y)) \equiv 0 \pmod{7}$ for every prime $p \notin \{2, 7\}$, then*

$$\frac{x - \theta y}{f'(\theta)} \in \mathbb{Q}^\times \cdot \{\nu^3 \sigma(\nu) : \nu \in K^\times\}.$$

(3) *Suppose that $F(x, y) = kw^7$ with $w \in \mathbb{Z}$, for one of the pairs (F, k) of Table 3 with $7 \mid k$ (so that $7 \mid F(x, y)$), and let $\delta \in \Delta(\theta, k)$ and $\omega \in K^\times$ be such that $x - \theta y = \delta \omega^7$, as in Proposition 3.3. Then $\kappa(\delta/f'(\theta)) = 0$, and $\delta/f'(\theta) = r\nu^3 \sigma(\nu)$ for some $r \in \mathbb{Q}^\times$ and $\nu \in K^\times$. In particular, $[v] \mapsto [\nu v]$ is an isomorphism from $Z_{\theta, \delta}$ to Z_0 over $\mathbb{Q}$, compatible with the maps to $\mathbb{P}^1$, and $Z_{\theta, \delta}$ is isomorphic to the Klein quartic $\mathcal{Z}$ over $\mathbb{Q}$.*

Proof. We prove (1). Suppose first that $\bullet = \text{sp}$. By Lemma 2.1(2b), $x + y = 7s$ for some integer s , and $7 \nmid y$. By (5.1),

$$x - \theta y = -y\varpi + 7s = \varpi(-y + s\theta^{-2}\varpi^2),$$

so $\kappa(x - \theta y) = 0 + 4 = 4$. On the other hand $f'(\theta) = \varpi^2(3 - 2\varpi/\theta)$ by (5.1), and $3 - 2\varpi/\theta \equiv 3 + 2\varpi \equiv 3(1 + 3\varpi) \pmod{\varpi^2}$, since $\theta \equiv -1$ and $2 \cdot 3^{-1} = 3$ in $\mathbb{F}_7$, so $\kappa(f'(\theta)) = 3 + 8 = 4$ as well. If $\bullet = \text{ns}$, then $7 \mid x$ and $7 \nmid y$, and by (1.4) and (1.5),

$$\frac{x - \theta_{\text{ns}} y}{f'_{\text{ns}}(\theta_{\text{ns}})} = \frac{1}{4} \cdot \frac{x' - \theta_{\text{sp}} y'}{f'_{\text{sp}}(\theta_{\text{sp}})}, \quad (x', y') = (x - 5y, -2y),$$

with $7 \mid x' + y'$ and $7 \nmid y'$; the computation above applies to (x', y') , and $\kappa(4) = 0$.

For (2), let γ be the element in question. The argument in the proof of Proposition 3.3 shows that $x - \theta y \in \mathcal{O}_{K, \mathcal{S}}^\times K^{\times 7}$ with $\mathcal{S} = \{2, 7\}$. Since 2 is inert, $\mathcal{O}_{K, \mathcal{S}}^\times = 2^\mathbb{Z} \times \mathcal{O}_K[1/7]^\times$, and $f'(\theta) \in \mathbb{Q}^\times \mathcal{O}_K[1/7]^\times$ by (1.5); hence the class of γ in $K^\times / \mathbb{Q}^\times K^{\times 7}$ lies in Σ . By Lemma 5.2(4) this class lies in $(3 + \sigma)\Sigma$, and by Lemma 5.2(2) it is therefore the class of $u^3 \sigma(u)$ for a unit u . This means that $\gamma = r u^3 \sigma(u) \lambda^7$ with $r \in \mathbb{Q}^\times$, $u \in \mathcal{O}_K^\times$ and $\lambda \in K^\times$. We conclude with (4.6).

For (3), we have $7 \mid F(x, y)$ because $7 \mid k$, and $v_p(F(x, y)) = 7v_p(w)$ for every prime $p \notin \{2, 7\}$ because k divides 56. Since κ is trivial on seventh powers (Lemma 5.1), (1) gives

$$\kappa\left(\frac{\delta}{f'(\theta)}\right) = \kappa\left(\frac{x - \theta y}{f'(\theta)}\right) - 7\kappa(\omega) = 0.$$

In other words, the 7-adic condition depends only on the class of $x - \theta y$ in $K^\times/K^{\times 7}$, which is the class of δ ; so it is a condition on $\delta \in \Delta(\theta, k)$. By (2) and (4.6), $\delta/f'(\theta) = ((x - \theta y)/f'(\theta))\omega^{-7}$ lies in $\mathbb{Q}^\times \cdot \{\nu^3\sigma(\nu) : \nu \in K^\times\}$. Now Lemma 4.3, applied to δ and $f'(\theta)$ in place of δ' and δ , gives the isomorphism $Z_{\theta, \delta} \rightarrow Z_{\theta, f'(\theta)} = Z_0$, and Corollary 4.7 gives $Z_0 \cong \mathcal{Z}$. $\square$

Proof of Theorem 1. We first take $(F, k) = (F_{\text{sp}}, 7)$, so that $\theta = \theta_{\text{sp}}$ as in the rest of this section. Let (x, y, w) be a solution of (1.3). By Proposition 3.3, $x - \theta y = \delta\omega^7$ with $\delta \in \Delta(\theta, k)$, and $[v] := \psi([\omega])$ is a rational point of $Z_{\theta, \delta}$ with $\varphi_{\theta, \delta}([v]) = (x : y)$, by Lemma 4.1. Since $7 \mid k$, Proposition 5.3(3) gives $\delta/f'(\theta) = r\nu^3\sigma(\nu)$ with $r \in \mathbb{Q}^\times$ and $\nu \in K^\times$, and the isomorphism $[v] \mapsto [\nu v]$ from $Z_{\theta, \delta}$ to Z_0 , compatible with the maps to $\mathbb{P}^1$. By Corollary 4.7, $\nu v \in \mathbb{Q}^\times \sigma^i(\alpha)$ for some $i \in \{0, 1, 2\}$, and therefore

$$x - \theta y \in \mathbb{Q}^\times \cdot f'(\theta) \sigma^i(\alpha^3 \sigma(\alpha)).$$

For $\theta = \theta_{\text{sp}}$ one computes

$$(5.2) \quad f'(\theta) \sigma^i(\alpha^3 \sigma(\alpha)) = -7^3(1 + \theta), \quad 7^3(5 - 2\theta), \quad -7^3(4 - 3\theta) \quad (i = 0, 1, 2).$$

Hence $(x, y) = \pm(1, -1)$, $\pm(5, 2)$ or $\pm(4, 3)$, where F_{sp} takes the values ± 7 . This proves the theorem for $(F_{\text{sp}}, 7)$.

For F_{ns} we use Lemma 2.1(4). Let $g = \gcd(x - 5y, 2y) \in \{1, 2\}$; it equals 2 exactly when x and y are both odd. Then $(x', y') = (x - 5y, -2y)/g$ is a primitive pair with $g^3 F_{\text{sp}}(x', y') = kw^7$. Since $F_{\text{sp}}(x', y')$ is odd by Lemma 2.1(3), and $v_2(k) \in \{0, 3\}$, w is odd, $g = 1$ if $k = 7$ and $g = 2$ if $k = 56$, and $F_{\text{sp}}(x', y') = 7w^7$ in both cases. By the split case, (x', y') is one of $\pm(1, -1)$, $\pm(5, 2)$, $\pm(4, 3)$. If $g = 1$ then y' is even, so $(x', y') = \pm(5, 2)$ and $(x, y) = \pm(0, 1)$. If $g = 2$ then $y = -y'$ and $x = 2x' - 5y'$ with y' odd, so $(x, y) = \pm(7, 1)$ or $\pm(7, 3)$.

Finally, $j_{\text{ns}}(0) = j_{\text{sp}}(-1) = 0$, while the j -invariants of the other four points, listed in Table 4, are not among the thirteen rational CM j -invariants. $\square$

Example 5.4. We follow the solution $(x, y, w) = (5, 2, -1)$ of $F_{\text{sp}}(x, y) = 7w^7$ through the proof. Here $x - \theta y = 5 - 2\theta$ has norm -7 , so it lies in $\mathcal{O}_K[1/7]^\times$, and in Proposition 3.3 we may take $\delta = 5 - 2\theta$ and $\omega = 1$; thus $[1] \in C_{\theta, \delta}(\mathbb{Q})$, and $\eta = \delta/f'(\theta)$. With $\varpi = 1 + \theta$ and $7 = \varpi^3\theta^{-2}$,

$$5 - 2\theta = 7 - 2\varpi = \varpi(-2 + \theta^{-2}\varpi^2),$$

so $\kappa(5 - 2\theta) = 0 + 4 = 4 = \kappa(f'(\theta))$ and $\kappa(\eta) = 0$, as Proposition 5.3 predicts. Indeed, by (5.2) with $i = 1$,

$$\eta = \frac{5 - 2\theta}{f'(\theta)} = 7^{-3} \sigma(\alpha)^3 \sigma^2(\alpha) = r \nu^3 \sigma(\nu), \quad r = 7^{-3}, \quad \nu = \sigma(\alpha).$$

The point $[1] = \psi([1]) \in Z_{\theta, \delta}(\mathbb{Q})$ is carried by the isomorphism $[v] \mapsto [\nu v]$ of Lemma 4.3 to $[\sigma(\alpha)] \in Z_0(\mathbb{Q})$, which is the point $(0 : 1 : 0)$ of the Klein quartic under Proposition 4.4. The

other two solutions of $F_{\text{sp}}(x, y) = \pm 7$ correspond to $\sigma(\eta)$ and $\sigma^2(\eta)$, that is, to $i = 2$ and $i = 0$ in (5.2).

Remark 5.5. The curves $C_{\theta, \delta}$ are not strictly needed for this proof: by Proposition 5.3 one can pass directly from a solution to a rational point of Z_0 . We have kept them because they are the natural objects of the descent, and because ψ explains where $Z_{\theta, \delta}$ comes from. For each of the six pairs (F, k) , the 7 classes $\delta \in \Delta(\theta, k)$ with $\kappa(\delta/f'(\theta)) = 0$ give the seven coverings of $Z_0 \cong \mathcal{Z}$ by twists of $\mathcal{F}$, three of which have rational points. By Proposition 5.6 below, these are the only $\delta \in \Delta(\theta, k)$ for which $Z_{\theta, \delta}$ is isomorphic to $\mathcal{Z}$ over $\mathbb{Q}$.

For $\eta \in K^\times$ we write $Z(\eta) := \{[v] \in \mathbb{P}(\mathcal{V}_K) : \text{Tr}(\eta v^3 \sigma(v)) = 0\}$ for the plane quartic of Remark 3.1, so that $Z_{\theta, \delta} = Z(\delta/f'(\theta))$ for both roots θ , and $Z_0 = Z(1)$. The following proposition, which is not needed for the proof of Theorem 1, shows that Proposition 5.3(3) cannot be improved: the 7-adic condition is exactly the condition for $Z_{\theta, \delta}$ to be the Klein quartic.

Proposition 5.6. *Let (F, k) be one of the six pairs of Table 3, let θ be the corresponding root, let $\delta \in \Delta(\theta, k)$, and let $\beta := \varepsilon_1 \varepsilon_2^4$.*

- (1) *The curve $Z_{\theta, \delta}$ is isomorphic to the Klein quartic $\mathcal{Z}$ over $\mathbb{Q}$ if and only if $\kappa(\delta/f'(\theta)) = 0$. This holds for exactly 7 of the 49 elements of $\Delta(\theta, k)$.*
- (2) *If $\kappa(\delta/f'(\theta)) \in \{1, 2, 4\}$, then $Z_{\theta, \delta}$ is isomorphic to $Z(\beta)$ over $\mathbb{Q}$, and if $\kappa(\delta/f'(\theta)) \in \{3, 5, 6\}$, then it is isomorphic to $Z(\beta^3)$. The three curves $\mathcal{Z}$, $Z(\beta)$ and $Z(\beta^3)$ are pairwise non-isomorphic over $\mathbb{Q}$.*

Proof. Put $\eta := \delta/f'(\theta)$. By Lemma 3.6(2), the class of η lies in Σ , and $\delta \mapsto [\eta]$ is a bijection from $\Delta(\theta, k)$ to Σ . By Lemma 5.2(3), we can write $[\eta] = s[\varepsilon_1 \varepsilon_2^2] + c[\beta]$ in additive notation, with $s, c \in \mathbb{F}_7$, and then $\kappa(\eta) = 4c$, since $\kappa(\varepsilon_1 \varepsilon_2^2) = 0$ and $\kappa(\beta) = 4$. In particular $\kappa(\eta) = 0$ for exactly 7 of the 49 classes. Since $[\varepsilon_1 \varepsilon_2^2]$ lies in $\Sigma_2 = (3 + \sigma)\Sigma$, the argument of the proof of Proposition 5.3(2) gives $\eta = r \nu^3 \sigma(\nu) \beta^c$ with $r \in \mathbb{Q}^\times$ and $\nu \in K^\times$, so $Z_{\theta, \delta} = Z(\eta) \cong Z(\beta^c)$ over $\mathbb{Q}$, by Lemma 4.3. If $c = 0$, this is $Z(1) = Z_0 \cong \mathcal{Z}$, by Corollary 4.7.

Next, for every $\eta' \in K^\times$ the $\mathbb{Q}$ -linear automorphism $v \mapsto \sigma(v)$ of $\mathcal{V}_K$ induces an isomorphism $Z(\eta') \rightarrow Z(\sigma(\eta'))$ over $\mathbb{Q}$, because $\text{Tr}(\sigma(\eta') \sigma(v)^3 \sigma^2(v)) = \text{Tr}(\eta' v^3 \sigma(v))$. By Lemma 5.2(3), $\sigma[\beta] = 4[\beta]$, so $\sigma(\beta^c)$ and β^{4c} have the same class in Σ ; hence $Z(\beta^c) \cong Z(\sigma(\beta^c)) \cong Z(\beta^{4c})$, the second isomorphism by (4.6) and Lemma 4.3. Since 4 has order 3 in $\mathbb{F}_7^\times$, the curves $Z(\beta^c)$ with $c \in \{1, 4, 2\}$ are isomorphic to $Z(\beta)$, and those with $c \in \{3, 5, 6\}$ to $Z(\beta^3)$; these values of c correspond to $\kappa(\eta) = 4c \in \{4, 2, 1\}$ and $\{5, 6, 3\}$ respectively.

It remains to show that $Z(1)$, $Z(\beta)$ and $Z(\beta^3)$ are pairwise non-isomorphic. In the coordinates $\mathbf{a}_0 + \mathbf{a}_1 \theta_{\text{sp}} + \mathbf{a}_2 \theta_{\text{sp}}^2$, the curve $Z(\eta')$ with $\eta' \in \mathcal{O}_K$ is defined by the form $\text{Tr}(\eta' u_0^3 u_1) \in \mathbb{Z}[\mathbf{a}_0, \mathbf{a}_1, \mathbf{a}_2]$; divide it by the greatest common divisor of its coefficients. A computation in Magma shows that for $\eta' \in \{1, \beta, \beta^3\}$ these three primitive quartic forms define smooth curves over $\mathbb{F}_{29}$, with 24, 45 and 17 points over $\mathbb{F}_{29}$ respectively. So the three curves have good reduction at 29. An isomorphism over $\mathbb{Q}_{29}$ between the generic fibers of two smooth proper curves of genus at least 2 over $\mathbb{Z}_{29}$ extends to an isomorphism over $\mathbb{Z}_{29}$, because the scheme of isomorphisms between them is finite, hence proper, over $\mathbb{Z}_{29}$ [DM69, Theorem 1.11]; reducing it modulo 29 gives an isomorphism of the reductions, which is impossible since they have different numbers of points. This proves (2), and (1) follows: $Z_{\theta, \delta} \cong \mathcal{Z}$ forces $c = 0$, that is, $\kappa(\eta) = 0$. $\square$

Remark 5.7. The isomorphism $Z_{\theta,\delta} \cong Z_0$ for $\kappa(\delta/f'(\theta)) = 0$ is compatible with the maps φ to $\mathbb{P}^1$, but the isomorphisms $Z(\beta^c) \cong Z(\beta^{4c})$ induced by σ are not: they carry $\varphi_{\theta,\delta}$ to $[v] \mapsto [\sigma(\delta) v^3 \sigma(v)]$, whose target is $\mathbb{P}(\mathbb{Q} + \mathbb{Q}\sigma(\theta))$. So Proposition 5.6 classifies the quartics, not the coverings. The rational points of $Z(\beta)$ and $Z(\beta^3)$ are not determined in this paper. The three curves have good reduction and the same number of points over $\mathbb{F}_p$ for every prime $p < 29$ other than 7, so 29 is the smallest prime that can be used in the proof. (It is also the smallest prime $p \equiv 1 \pmod{7}$.)

6. PROOF OF THEOREM 2

In this section we show that the eight non-CM points of $X_{\text{ns}}^+(7)$ and $X_{\text{sp}}^+(7)$ resulting from Theorem 1 and [FL26] do not lift to rational points of $X_{\text{ns}}^\sharp(49)$ or $X_{\text{sp}}^\sharp(49)$.

6.1. A Frobenius criterion.

Lemma 6.1. *Let $\bullet \in \{\text{ns}, \text{sp}\}$, and let $g \in G^\sharp$ be such that $7 \nmid \text{tr}(g)(\text{tr}(g)^2 - 4 \det g)$. Then g^{48} is a scalar matrix.*

Proof. Let $\bar{g} \in C^+$ be the reduction of g . The elements of C^+ outside the Cartan subgroup C have trace zero, so $\bar{g} \in C$; and $\bar{g}$ is not scalar, because its characteristic polynomial is separable. Hence the centralizer of $\bar{g}$ in $M_2(\mathbb{F}_7)$ is $\mathbb{F}_7[\bar{g}]$, a subalgebra of dimension 2 of $\mathfrak{c}$, that is, $\mathfrak{c}$ itself. Since $\bar{g}^{48} = 1$, we have $g^{48} = I + 7A$ with $A \in M_2(\mathbb{F}_7)$; this element commutes with g , so $A \in \mathfrak{c}$, and it lies in $G^\sharp$, so $A \in V$. Finally, $V \cap \mathfrak{c}$ consists of the scalar matrices by Lemma 2.2(1). $\square$

Corollary 6.2. *Let E be an elliptic curve over $\mathbb{Q}$ such that $\rho_{E,49}(\text{Gal}_{\mathbb{Q}})$ is conjugate to a subgroup of $G_{\text{ns}}^\sharp(49)$ or of $G_{\text{sp}}^\sharp(49)$. Then for every prime $p \neq 7$ of good reduction such that $7 \nmid a_p(E)(a_p(E)^2 - 4p)$, the matrix*

$$\begin{pmatrix} 0 & -p \\ 1 & a_p(E) \end{pmatrix}^{48} \in M_2(\mathbb{Z}/49)$$

is scalar. This condition is invariant under quadratic twists of E .

Proof. The matrix $g = \rho_{E,49}(\text{Frob}_p)$ has characteristic polynomial $t^2 - a_p(E)t + p$, which is separable modulo 7. So g has a cyclic vector modulo 7, hence (by Nakayama's lemma) over $\mathbb{Z}/49$, and it is conjugate to the companion matrix of its characteristic polynomial. Now apply Lemma 6.1. A quadratic twist replaces $a_p(E)$ by $\pm a_p(E)$, and the companion matrix by a conjugate of its negative. $\square$

The criterion is far from vacuous: in the full preimage of $C_{\text{ns}}^+(7)$ in $\text{GL}_2(\mathbb{Z}/49)$, and likewise in that of $C_{\text{sp}}^+(7)$, exactly 6/7 of the elements g with $7 \nmid \text{tr}(g)(\text{tr}(g)^2 - 4 \det g)$ violate it. For instance, an elliptic curve with the j -invariant of the point $t = 5/2$ of Table 4 has $a_{11} = \pm 2$, and

$$\begin{pmatrix} 0 & -11 \\ 1 & 2 \end{pmatrix}^{48} \equiv \begin{pmatrix} 43 & 42 \\ 14 & 22 \end{pmatrix} \pmod{49}$$

is not scalar.

6.2. The candidates. Table 4 lists the non-CM points that occur in the proof. For each of them we give the first prime p at which the criterion of Corollary 6.2 fails, together with $|a_p|$, for an elliptic curve with the given j -invariant and good reduction at p . Such a curve exists: in every row j is p -integral and $j \not\equiv 0, 1728 \pmod{p}$, so the reduction has automorphism group $\{\pm 1\}$, inertia at p acts through it, and some quadratic twist has good reduction at p . Its a_p is determined up to sign by j , and by Corollary 6.2 the outcome does not depend on the sign.

curve	t	j	(p, a_p)
$X_{\text{ns}}^+(7)$	7	$2^3 \cdot 5^3 \cdot 7^5$	(3, 1)
$X_{\text{ns}}^+(7)$	7/3	$2^{15} \cdot 7^5$	(3, 2)
$X_{\text{sp}}^+(7)$	5/2	$3^3 \cdot 5 \cdot 7^5 / 2^7$	(11, 2)
$X_{\text{sp}}^+(7)$	4/3	$-2^8 \cdot 5^3 \cdot 7^5 \cdot 37^3 / 3^7$	(13, 2)
$X_{\text{sp}}^+(7)$	3/2	$3 \cdot 5^3 \cdot 11^3 \cdot 17^3 \cdot 43^3 / 2^7$	(23, 4)
$X_{\text{sp}}^+(7)$	-1/4	$3^3 \cdot 37^3 \cdot 149^3 \cdot 2389^3 / 2^{14}$	(5, 1)
$X_{\text{sp}}^+(7)$	13/9	$-2^{18} \cdot 5^3 \cdot 11^3 \cdot 13 \cdot 29^3 \cdot 37^3 \cdot 67^3 \cdot 127^3 / 3^{14}$	(17, 3)
$X_{\text{sp}}^+(7)$	14/5	$2^4 \cdot 3^3 \cdot 7^4 \cdot 19^3 \cdot 23^3 \cdot 43^3 \cdot 163^3 / 5^7$	(31, 10)

TABLE 4. Non-CM candidates, and a prime at which Corollary 6.2 fails. The first four rows come from Theorem 1; the last four from the solutions of $F_{\text{sp}}(x, y) = \pm 1$.

Proof of Theorem 2. Let X be one of the two curves and G the corresponding group. Let $P \in X(\mathbb{Q})$ be a non-cuspidal point which is not CM. Then $j(P) \notin \{0, 1728\}$, and since $-I \in G$ there is an elliptic curve E over $\mathbb{Q}$ with $j(E) = j(P)$, without CM, such that $\rho_{E,49}(\text{Gal}_{\mathbb{Q}})$ is conjugate to a subgroup of G ; see [RSZB22, Section 2.3]. Since $-I \in G$, the same holds for every quadratic twist of E ; when we apply Corollary 6.2 at a prime p below, we take E to be a twist with good reduction at p , as explained before Table 4. Proposition 2.8 provides coprime integers x, y and an integer w with $j(E) = j_{\bullet}(x/y)$ and $F_{\bullet}(x, y) = kw^7$.

If $7 \mid k$, then by Theorem 1 the point $t = x/y$ is one of $0, 7, 7/3$ (non-split case) or $-1, 5/2, 4/3$ (split case). Since $j_{\text{ns}}(0) = j_{\text{sp}}(-1) = 0$, the point t is one of the first four entries of Table 4, and Corollary 6.2 gives a contradiction.

If $7 \nmid k$, then $X = X_{\text{sp}}^{\sharp}(49)$ and $k = 1$. By [FL26, Corollary 3.3(1), Theorem 3.5(1)] (see the proof of [FL26, Corollary 3.7]), $w = \pm 1$, so (x, y) is a solution of the Thue equation $F_{\text{sp}}(x, y) = \pm 1$. Its solutions, up to sign, are

$$(6.1) \quad (1, 0), (0, 1), (1, 1), (2, 1), (3, 1), (3, 2), (1, -4), (13, 9), (14, 5),$$

as one finds with the Thue solver of **Magma**. The points $t = \infty, 0, 1, 2, 3$ are the cusp and CM points with discriminants $-3, -19, -12, -27$. The remaining four are the last four entries of Table 4, and Corollary 6.2 gives a contradiction again.

It remains to show that a CM point $P \in X(\mathbb{Q})$ has $j(P) = 0$. Suppose first that $j(P) = 1728$. Then, since $-I \in G$, some elliptic curve E over $\mathbb{Q}$ with $j(E) = 1728$ has $\rho_{E,49}(\text{Gal}_{\mathbb{Q}})$ conjugate to a subgroup of G [RSZB22, Section 2.3, Remark 2.1, and Section 12.2, Proposition 12.1]. Such an E is the twist of $E_1: y^2 = x^3 - x$ by a character χ with values in $\text{Aut}(E_1) = \mathbb{Z}[i]^\times$, so that $\rho_{E,49} = \rho_{E_1,49}\chi$ on $\text{Gal}_{\mathbb{Q}(i)}$. Let $\tau \in \text{Gal}_{\mathbb{Q}(i)}$ be a Frobenius element at a prime above 5, which splits in $\mathbb{Z}[i]$. Both $\rho_{E_1,49}(\tau)$ and $\chi(\tau)$ lie in the Cartan subgroup $(\mathbb{Z}[i]/49)^\times$, so $g := \rho_{E,49}(\tau)$ is the image of $\pi\chi(\tau)$ for an element $\pi \in \mathbb{Z}[i]$ of norm 5, and its characteristic polynomial is $t^2 - at + 5$ with $a \in \{\pm 2, \pm 4\}$. Then $7 \nmid a(a^2 - 20)$, so Lemma 6.1 says that g^{48} is scalar; but g is conjugate to the companion matrix of $t^2 - at + 5$, as in the proof of Corollary 6.2, and for each of these values of a the 48th power of that matrix is not scalar modulo 49. So $j(P) \neq 1728$, and if $j(P) \neq 0$ then P comes from an elliptic curve E over $\mathbb{Q}$ as in the first paragraph of this proof, all of whose twists are quadratic. In the non-split case, [FL26, Lemma 2.11] applied to E shows that $7 \mid j(P)$, and 0 is the only one of the thirteen rational CM j -invariants that is divisible by 7. In the split case, write $j(P) = j_{\text{sp}}(x/y)$ with x, y coprime. Since $j(P)$ is an integer, Lemma 2.1(1) shows that $yF_{\text{sp}}(x, y)$ is, up to sign, a power of 7. If $7 \mid y$, then $7 \nmid xH_{\text{sp}}(x, y)$ and $7 \nmid F_{\text{sp}}(x, y)$, because $F_{\text{sp}} \equiv x^3$ and $H_{\text{sp}} \equiv x^9 \pmod{y}$, so $j(P) = xH_{\text{sp}}(x, y)^3/(yF_{\text{sp}}(x, y))^7$ is not an integer; hence $y = \pm 1$, and $F_{\text{sp}}(x, y) \in \{\pm 1, \pm 7\}$ by Lemma 2.1(2). The primitive solutions of these Thue equations are, up to sign, the nine pairs in (6.1) together with $(1, -1)$, $(5, 2)$ and $(4, 3)$; the CM points among these twelve are $t = 0, -1$, where $j = 0$, and $t = 1, 2, 3$, with discriminants $-19, -12, -27$. For $t \in \{1, 2, 3\}$ the criterion of Corollary 6.2 fails at $(p, |a_p|) = (5, 1)$, $(13, 2)$ and $(13, 5)$ respectively. $\square$

Remark 6.3. In the non-split case, the part of the proof above that treats non-CM points uses [FL26, Lemma 2.11] (through Proposition 2.8) to force $7 \mid k$. This can be avoided. If $7 \nmid k$, then $k \in \{1, 8\}$ and [FL26, Theorem 3.5(1)] gives $w = \pm 1$, so $t = x/y$ is one of Kenku's twelve integral points of $X_{\text{ns}}^+(7)$ [Ken85], namely

$$t \in \{0, \infty, 1, -1, 2, 3, 5, -3/5, 7, 7/3, 11/2, 19/9\};$$

see also [Elk99, Section 4.3]. Eight of them are CM, two are in Table 4, and the remaining two, $t = 11/2$ and $t = 19/9$, with

$$j = 2^6 \cdot 11^3 \cdot 23^3 \cdot 149^3 \cdot 269^3 \quad \text{and} \quad j = 2^9 \cdot 17^6 \cdot 19^3 \cdot 29^3 \cdot 149^3,$$

violate the criterion of Corollary 6.2 at $(p, |a_p|) = (3, 1)$ and $(3, 2)$ respectively.

Remark 6.4. The first point in the split part of Table 4, $j = 3^3 \cdot 5 \cdot 7^5/2^7$ at $t = 5/2$, is the exceptional j -invariant of level 7 of [Zyw15, Theorem 1.5(iii)], the constant function J_1 there. The proof of [FL26, Corollary 3.7] excludes the candidates with Zywna's algorithm [Zyw22] instead of Corollary 6.2.

7. ORIGIN OF THE IDEAS AND USE OF AI

This section records how the ideas in this paper arose, and what AI models contributed to it (see the AI use disclosure in the introduction). All of the work has been checked by the authors.

Our starting point was the work of Furio and Lombardo [FL26], who, through a careful analysis of the ramification of the j -maps, reduced the difficult problem of determining the rational points on the genus 9 curves $X_{\text{ns}}^\sharp(49)$ and $X_{\text{sp}}^\sharp(49)$ to that of solving the superelliptic equations $F(x, y) = kw^7$ in integers.

Our approach was inspired by Kummer’s proof of Fermat’s Last Theorem for regular primes [Kum50], as presented by Conrad [Con]. It led us to the curves $C_{\theta, \delta}$, and to rediscover the work of Bruin [Bru02, Bru06], who uses the same approach to parametrize the solutions of certain spherical generalized Fermat equations. We first worked with the curves $C_{\theta, \delta}$ directly over $\mathbb{Q}$. We found a map from each of them to a twist of the Klein quartic by computing their zeta functions at several primes and matching the resulting L -factors with those of curves of genus 3, using the LMFDB [LMFDB26] among other resources.

We then gave our work to Claude Opus 5, which pointed out that over K the curves $C_{\theta, \delta}$ are twists of the Fermat septic (Lemma 3.5), using the intrinsic description $\mathbb{P}^2(\mathbb{Q}) = K^\times / \mathbb{Q}^\times$ of Remark 3.1. This is not logically necessary for the proof, but we found it a valuable conceptual clarification, and rewrote part of the paper from this perspective.

At this stage we still had to determine the rational points on 49 curves of genus 15. We knew that they admitted quotients of genus 3, and observed that those quotients were twists of the Klein quartic. An ad hoc local argument at 7 reduced the number of twists to around 7. From here, it was clear to us that the approach of Furio–Lombardo would lead to success, and we were in the process of working out those details.

Meanwhile, after a talk by the second author on our partial progress at the ChaBONNty conference in Bonn,ⁱ Filip Najman started exploring this problem independently. He kindly informed us that he was able to determine the integer solutions of the remaining superelliptic equations, and that everything reduced to the Klein quartic. This surprised us, since our computations suggested that some nontrivial twists remained.

Finally, we gave our partial work to Claude Fable 5.1, which produced the following simplifications.

- Since $\theta_{\text{ns}} = 5 - 2\theta_{\text{sp}}$, the two roots span, together with 1, the same plane $\mathbb{Q} + \mathbb{Q}\theta$ in K . So the two descents involve the same curves, and only the coordinate on the target changes (Remark 3.1, Lemma 3.6(3), Lemma 2.1(4)). This is not necessary for the result, but it saved us from duplicating our work.
- Our original version of the 7-adic step (Section 5.2) used an ad hoc local test that ruled out most of the classes δ ; based on our code Claude suggested Proposition 5.3 as a cleaner formulation of the method.
- We had misread [FL26, Corollary 3.3], and were duplicating some of their work with our own methods; Claude pointed out that several of the cases we were considering were already settled by that corollary.

Putting everything together, Najman turned out to be right: every twist that survives the 7-adic step is isomorphic over $\mathbb{Q}$ to the Klein quartic (Proposition 5.3(3)), and our results follow from there.

ⁱMax Planck Institute for Mathematics, June 29 to July 3, 2026; <https://www.mpim-bonn.mpg.de/ChaBONNty>.

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